CPSC 368
Databases in Data Science

Conceptual Database Design
The Entity-Relationship Model

Textbook Reference
Database Management Systems: Chapter 2

Databases: the continuing saga…

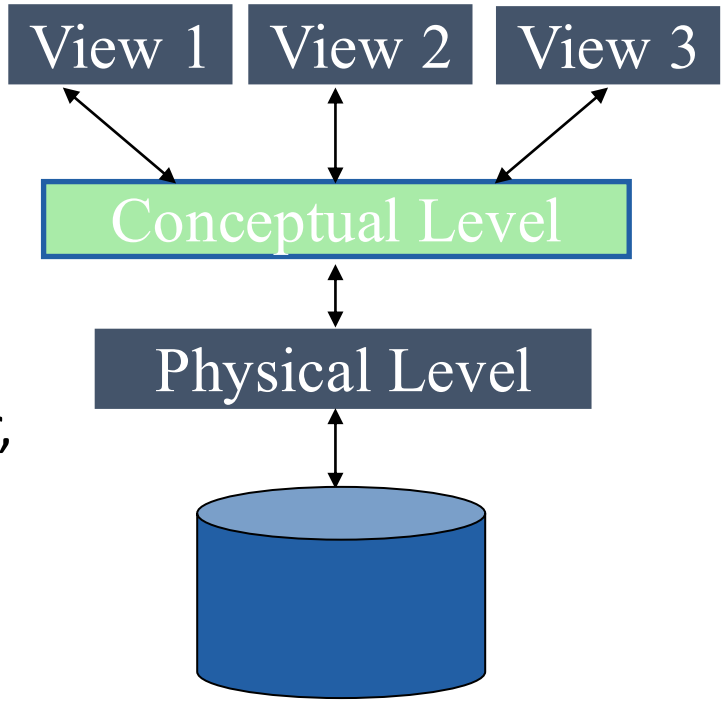
- We motivated that databases were great because they:
  - Store large amounts of data
  - Handle transactions
  - Allow efficient querying
  - *And many, many more classic favorites!*
- Before we can do all of these, we must design the database

Learning Goals

- Explain the purpose of an ER diagram, and list the major components.
- Given a problem description, create an ER diagram given a specification. Justify the decisions you make for entities, relationships, keys, key constraints, participation constraints, weak entities, is-a relationships, and aggregations.
- Given a problem description, identify alternative representations of the problem concepts and evaluate the choices
- Compare alternative ER models for the same domain and identify their strengths and weaknesses

Levels of Abstraction

- A major purpose of a DB system is to provide an abstract view of the data.
- Three abstraction levels:
  - **External (or View) level**: describes different part of the database to different users
    - Convenience, security, etc.
    - Compare views of student, registrar, & database admin.
  - **Conceptual (or Logical) level**: how data is perceived by the users
  - **Physical level**: how data is actually stored (404) – covers things like indexes, bits on disk



Schema and Instances

- We create the conceptual **schema** – the logical structure of the database (e.g., students take courses)

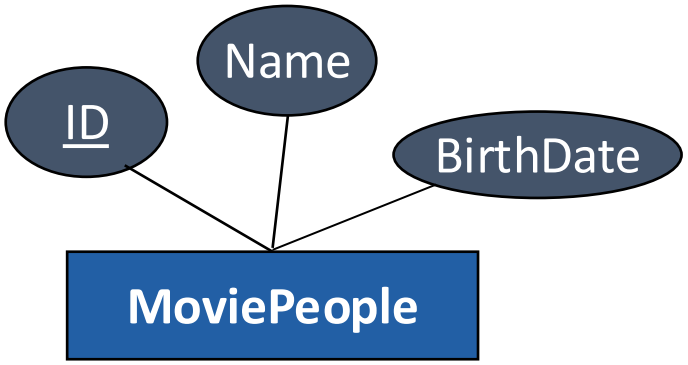
- Later we'll populate **instances** – the content of the database at a particular point in time
  - E.g., we've set the grading schema for the term but there are currently no grades for CPSC 368

Conceptual Database Design

- What are the entities and relationships in the enterprise?
  - Entities are usually nouns
    - e.g., Students and courses
  - Relationships are statements about 2 or more objects. Often, verbs.
    - e.g., a prof teaches a course
- What information about these entities and relationships should we store in the database?
- What integrity constraints or other rules hold?
- In relational databases, this is generally encoded in an **Entity-Relationship (ER) Diagram**

ER Model Basics: Entities

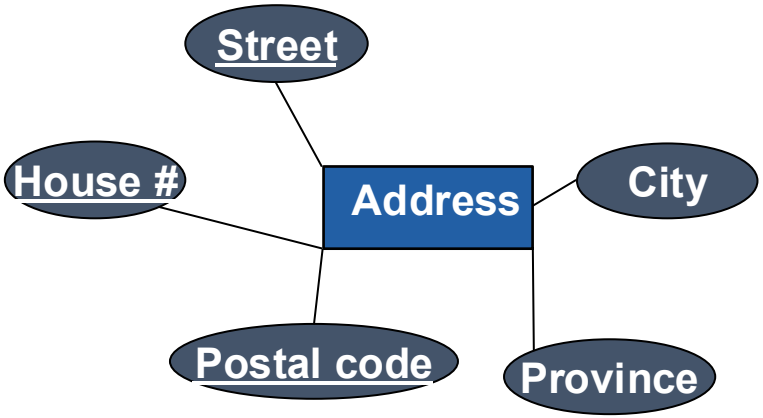
- ***Entity:*** Real-world object distinguishable from other objects.
  - An entity is described using a set of ***attributes***.



- ***Entity Set:*** A collection of similar entities. E.g., all Movie People.
  - All entities in an entity set have the same set of attributes. (Until we consider ISA hierarchies, anyway!)
  - Each attribute has a ***domain***. *(e.g.., float, date, int)*
  - Each entity set has a ***key***. The key is composed of all underlined attributes

Keys

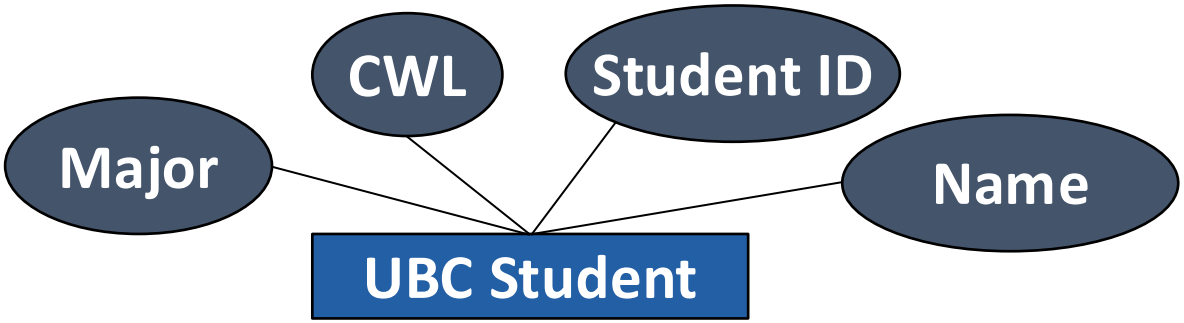
- Distinguish entities
- A *key* is the **minimal** set of one or more attributes which, taken collectively, identify uniquely an entity in an entity set.
- In Canada, ~50 addresses share the same postal code



- A *primary key* is the key chosen as the principal means to identify entities in an entity set
- The only keys shown in ER diagrams are primary keys (do not worry about this for now)

Minimal Key

- Every single attribute in the key is required to determine the values of the other attributes; no "unnecessary" attributes are included
- It does not say anything about the **number** of items in a key



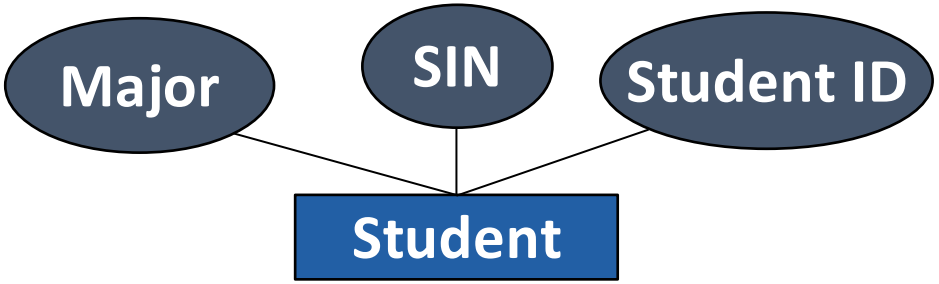
If there was a restriction such that each major can only have one person with a given name, would {major, name} be considered a minimal key?

A. Yes B. No

Keys: clicker question

Consider a student entity loosely based on the real world:

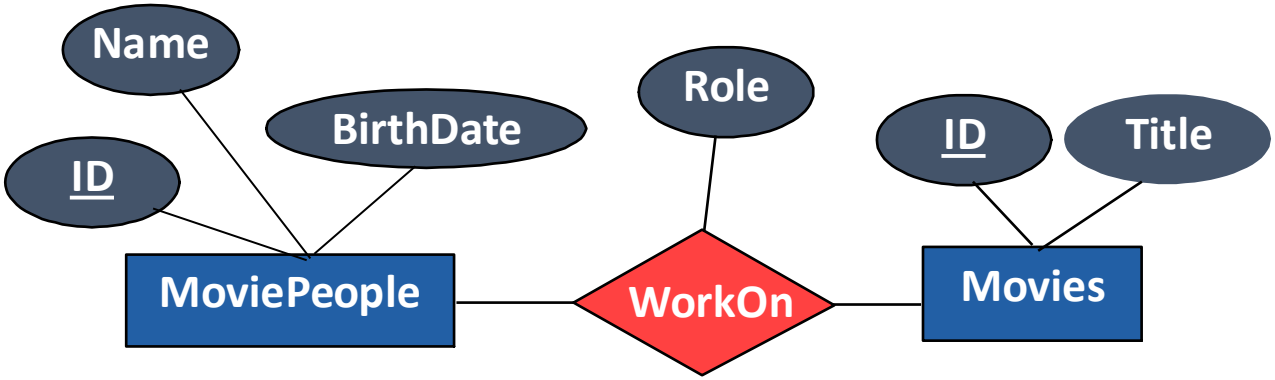
You may assume both SIN & Student ID are unique.



Which of the following is *not* a possible key for the student entity?

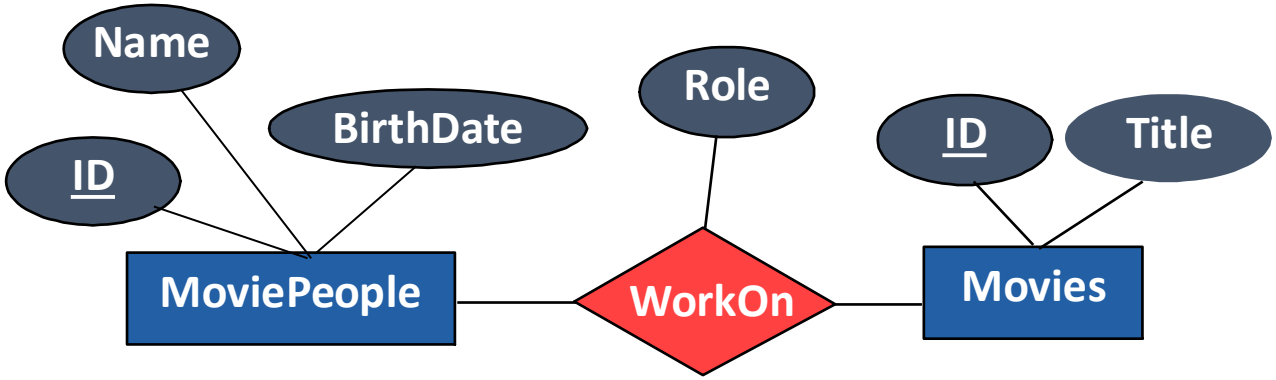
A. SIN & Student ID together
B. SIN
C. Student ID
D. All are possible
E. More than one is not possible

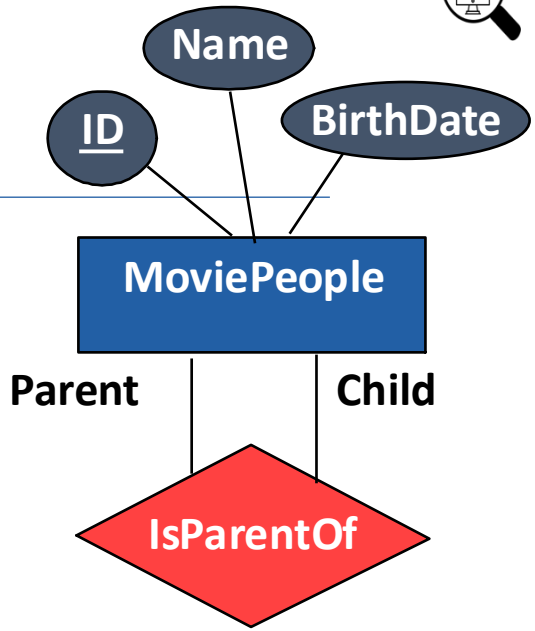
ER Model Basics (Cont.)



- ***Relationship:*** Association among two or more entities.
  - E.g., Michelle Yeoh worked on Everything Everywhere All at Once.

ER Model Basics (Cont.)

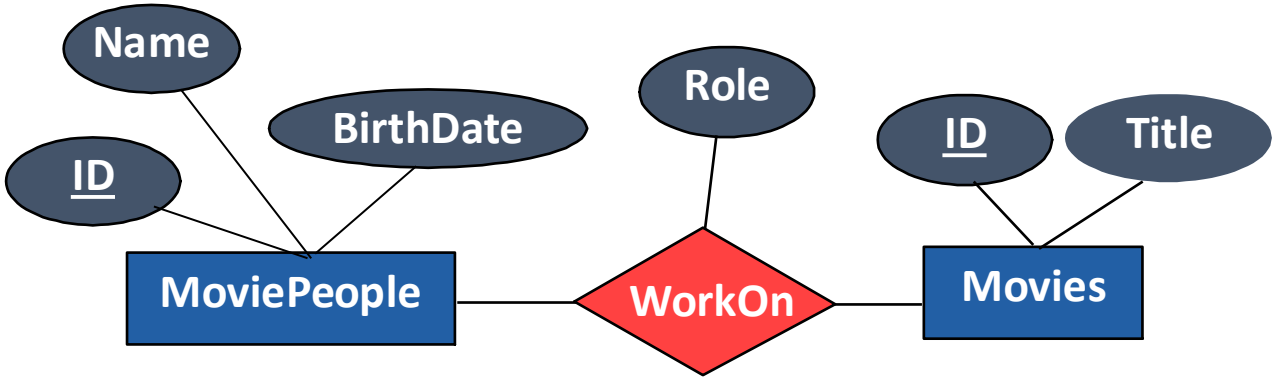


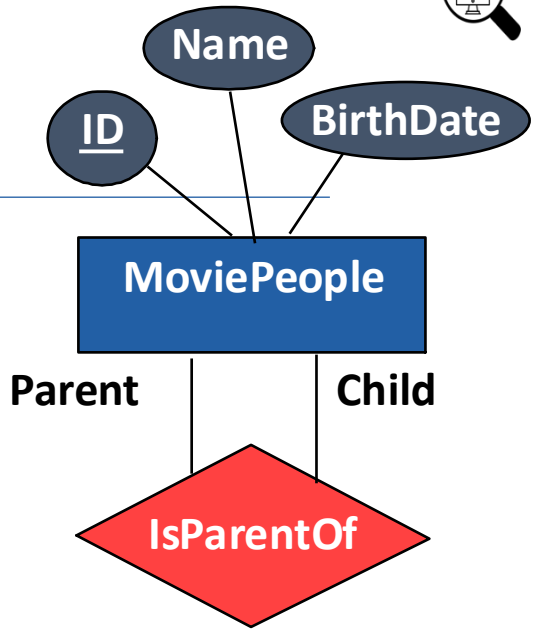


- ***Relationship Set:*** Collection of similar relationships.
  - Collection of all moviePeople that have worked in Movies.

- Same entity set could participate in different relationship sets, or in different "roles" in same set. (Kirk Douglas isParentOf Michael Douglas)

ER Model Basics (Cont.)





- A relationship must be uniquely defined by the entities involved (it may not be a key because it may not be minimal, as we'll see shortly)
- A relationship set may have ***descriptive attributes*** (like Role).
- An n-ary relationship set R relates n entity sets $E_1 \ldots E_n$; each relationship in R involves entities $e_1 \in E_1, \ldots, e_n \in E_n$
  - ***Degree*** or ***arity:*** # of entity sets in relationship (binary, ternary, etc.)

Exercise: Registrar’s database

- Design a registrar's database to store information about students, courses, the courses students have taken, and the grades students have gotten in these courses. Some relevant details are: Courses have a number, a department, and a title. for example, "CPSC100: Computational Thinking" has department = CPSC, number = 100, and title = “Computational Thinking."
- Numbers are assigned by departments, and different departments may use the same number.
- Students are represented by their (unique) student ID and their name.
- "Enrollments" each consist of a course, a student who took that course, and the grade the student got in the course.
- You should draw one or more E/R diagrams that represent this database structure correctly.

Exercise: Registrar’s database

Which of the following might you find in a correct E/R diagram?

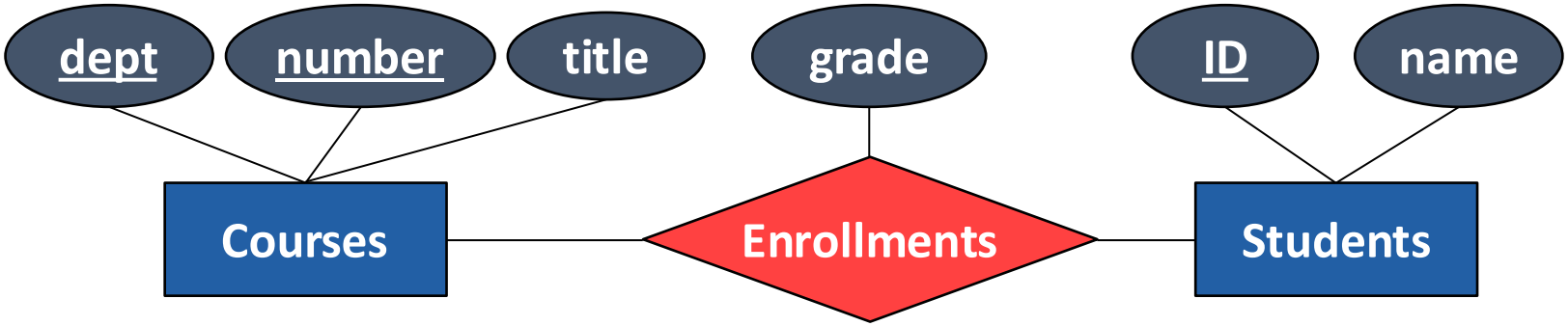
A. Entity set Students with attribute ID not underlined and name underlined.

B. Entity set Students with attributes ID and name underlined.

C. Entity set Courses with attributes department and number underlined and title not underlined.

D. Entity set Courses with attribute department underlined and attributes number and title not underlined.

E. None of the above



Okay, that worked pretty well…

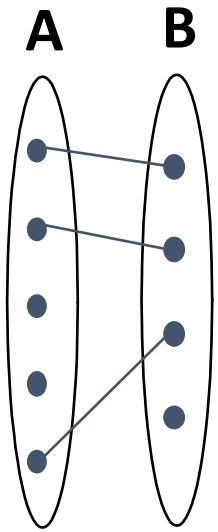
But what if we wanted to add instructors to courses and restrict that each could only have one instructor?

Cardinalities

- A **cardinality ratio** for a relationship set specifies the number of relationships in the set that an entity can participate in.

Let R be a relationship set between sets A and B. R can have 1 of 4 cardinalities:

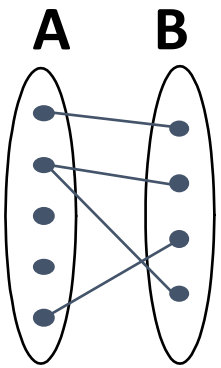
1. ***one-to-one*** from A to B:
   - an entity in A is associated with at most one entity in B and vice versa
   - e.g. A: student, B: student ID #



1-to-1

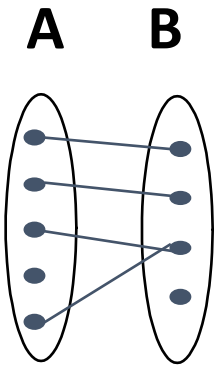
Cardinalities (cont')

2. ***one-to-many*** from A to B:
   - an entity in A is associated with any number of entities in B
   - an entity in B is associated with at most one entity in A
   - e.g. A: biological-mother, B: children



1-to Many

3. ***many-to-one*** *from A to B: switch A and B above*

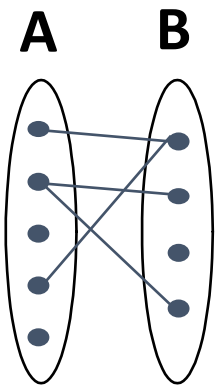


Many-to-1

Cardinalities (cont')

*4. **many-to-many** from A to B:*

- *an entity in A is associated with any number of entities in B and vice versa*
- *e.g. A: students, B: courses*

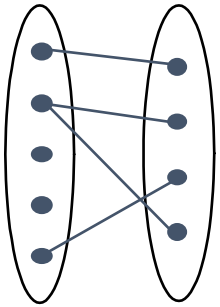


Many-to-Many

Cardinalities Exercise #1

What cardinality constraint does this diagram represent:

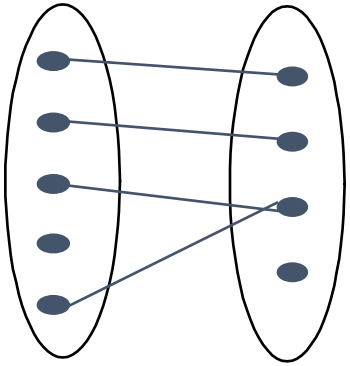
A. One-to-one
B. One-to-many
C. Many-to-one
D. Many-to-many



Cardinalities Exercise #2

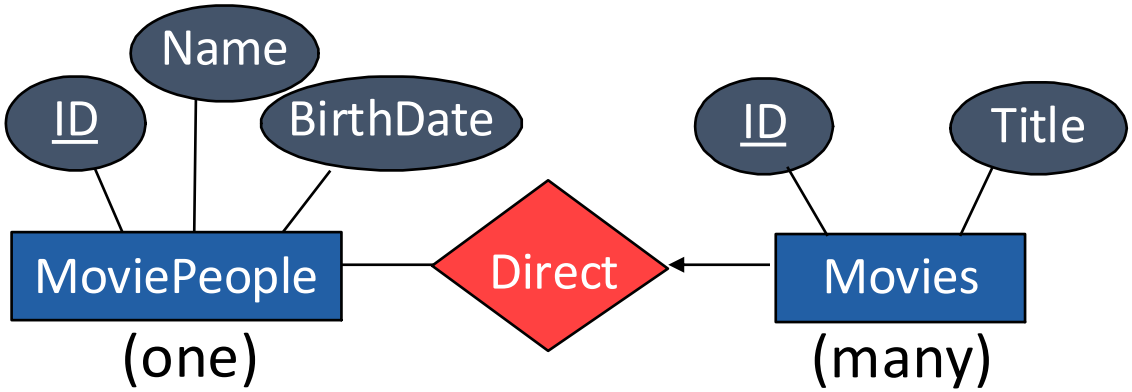
Which cardinality constraint is most appropriate for the relationship of students to overall course grades in CPSC 368?

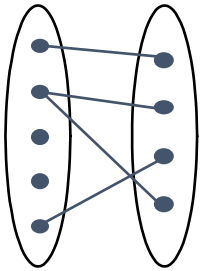
A. One-to-one
B. Many-to-one
C. One-to-many
D. Many-to-many



Key Constraints

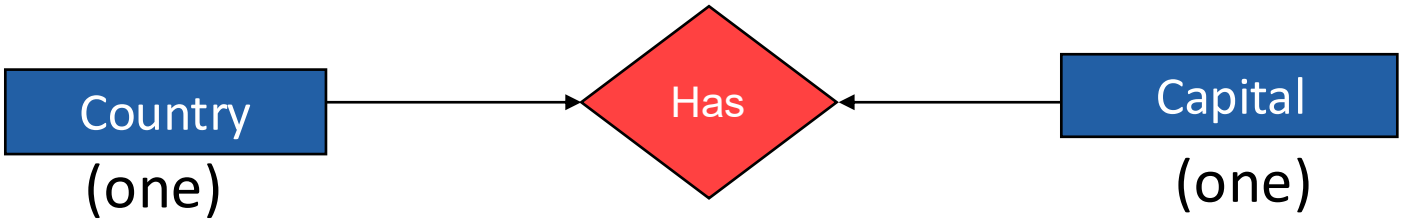
- The restriction imposed by a 1-to-1 and 1-to-many ratios are examples of ***key constraints***.
- A key constraint is shown with an arrow in the ER diagram.
- Important to consider when inserting data





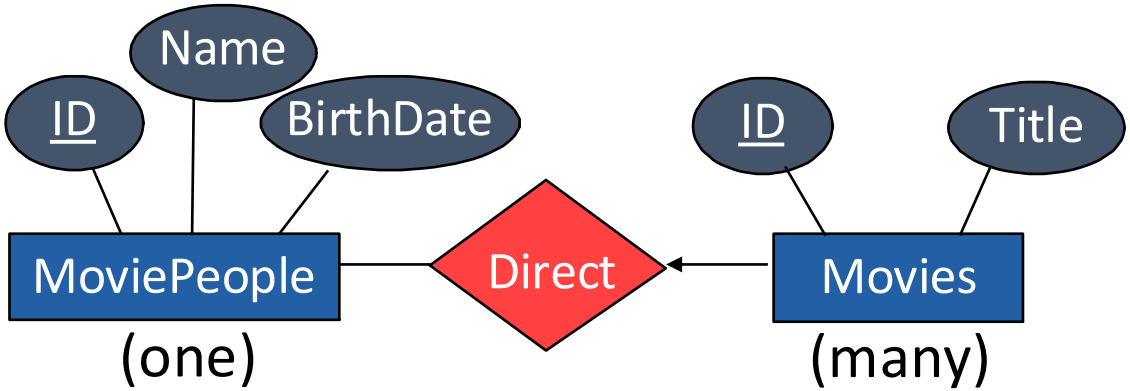
1-to Many

or

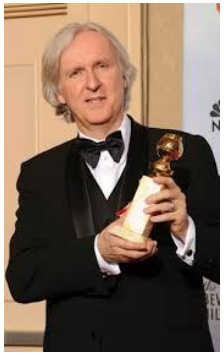


A brief digression on notation

The ER notation we use can be read: “if you know the entity with the arrow, then you know the relationship (and the other entities involved)” – the arrow points to the thing there is only one of



James Cameron

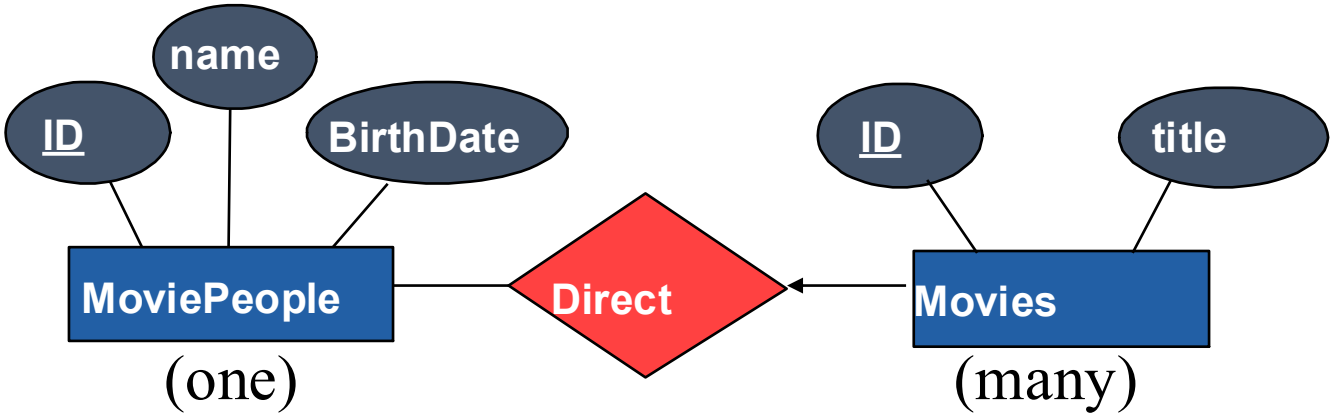


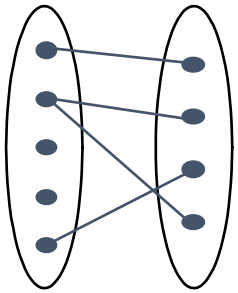
Clicker Exercise

In a one to many relationship like the one below, which statement is true?

A. If I know the movie, I definitely know who directed it

B. If I know the director, I definitely know which movie is being referred to in the Direct relationship





1-to Many

Line annotation example

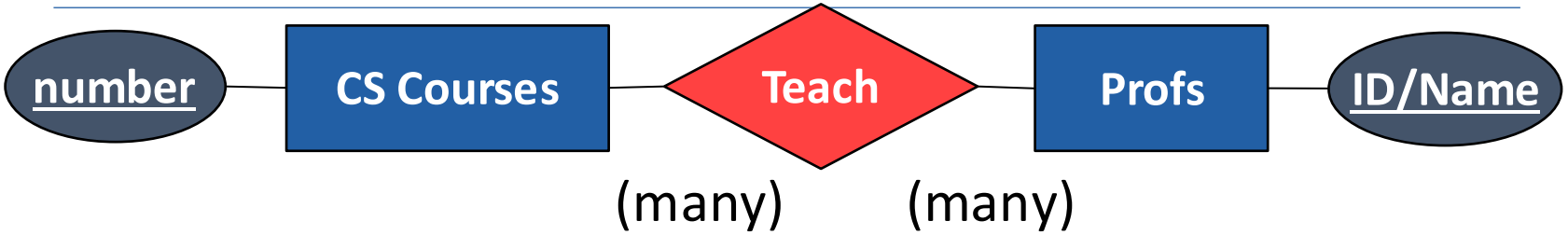
- Let's focus on a simplified version of "Profs teach courses":

- Let's assume that the following is our entire set of entities:

CS Courses Number
100
304
312
313
417

Profs ID/Name
Rachel
Steve
Jessica
Phong

Plain lines only



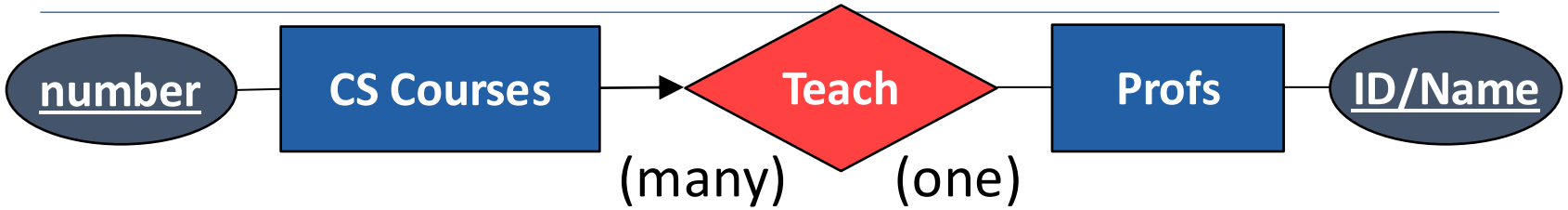
(many) (many)

CS Courses Number
100
304
312
313
417

Profs ID/Name
Rachel
Steve
Jessica
Phong

- Each course can be taught by more than one prof
- Each prof can teach more than one course
- Not all courses have to be taught by some prof
- Not all profs have to teach a course

Clicker question: Many-to-one. Which of the following would allow us to satisfy the given constraint?

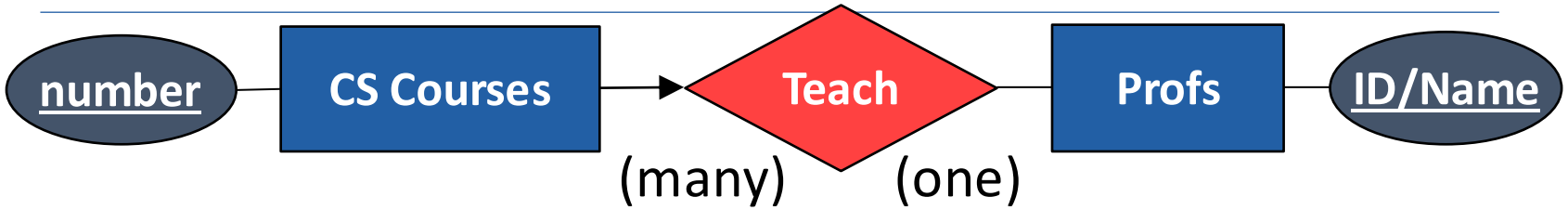


CS Courses Number
100
304
312
313
417

Profs ID/Name
Rachel
Steve
Jessica
Phong

A. Removing {(100,Rachel), (304, Jessica)} from Teach
B. Removing {(304,Rachel), (304, Jessica), (312, Steve)} from Teach
C. Adding {(417, Phong)} to Teach
D. All of the above
E. None of the above

Many to One



CS Courses Number
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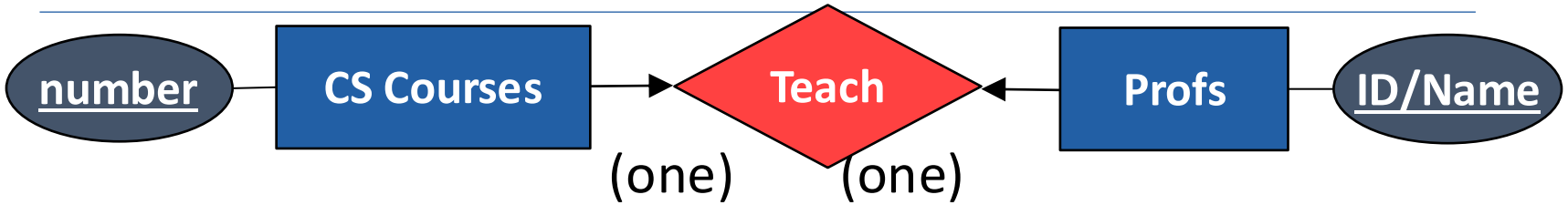
Profs ID/Name
Rachel
Steve
Jessica
Phong

- Each course can be taught by AT MOST ONE prof
- Each prof can teach zero or more courses
- Not all courses have to be taught by some prof
- Not all profs have to teach a course

One to One



CS Courses Number
100
304
312
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417

Profs ID/Name
Rachel
Steve
Jessica
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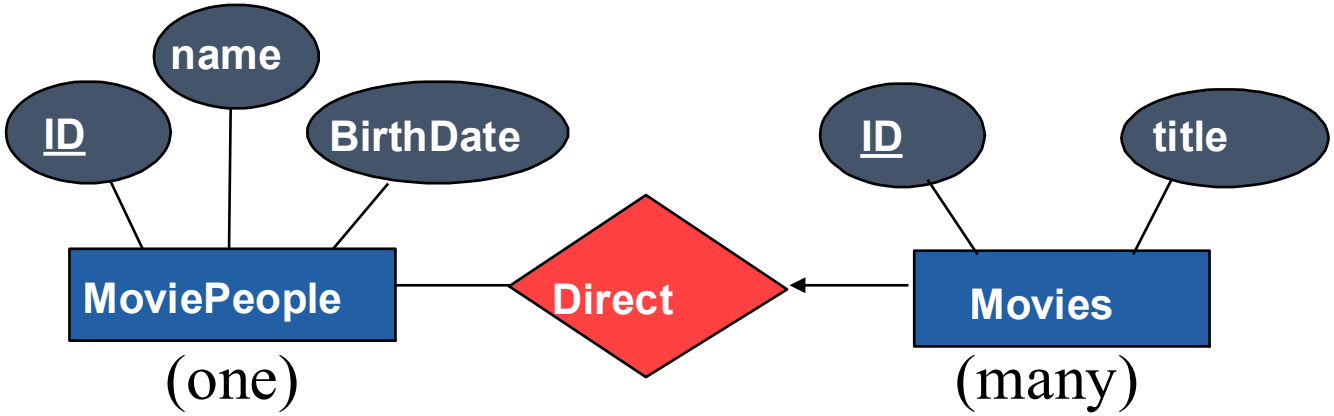
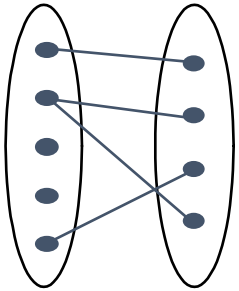
- Each course can be taught by AT MOST ONE prof
- Each prof can teach AT MOST ONE course
- Not all courses have to be taught by some prof
- Not all profs have to teach a course

Clicker Exercise Explained

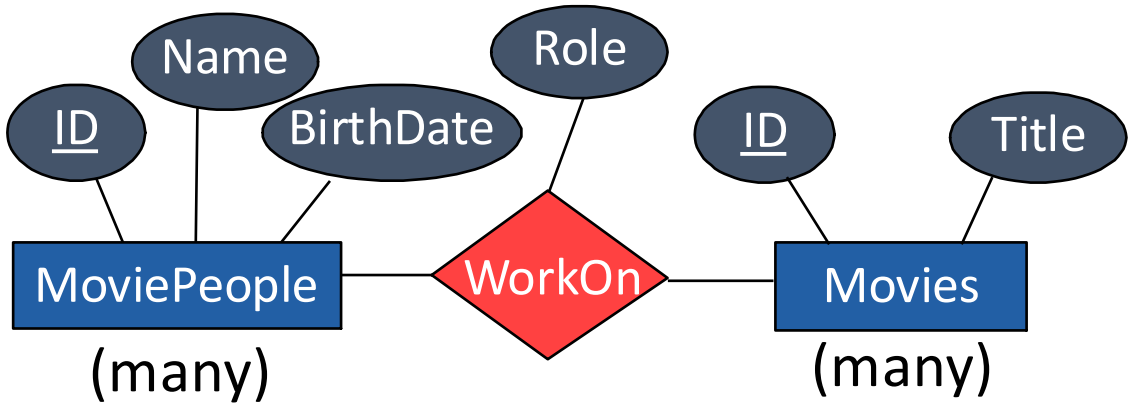
In other words, the key for a one to many relationship is the key of the entity that is on the many side!





1-to Many

How can we uniquely identify a relationship?



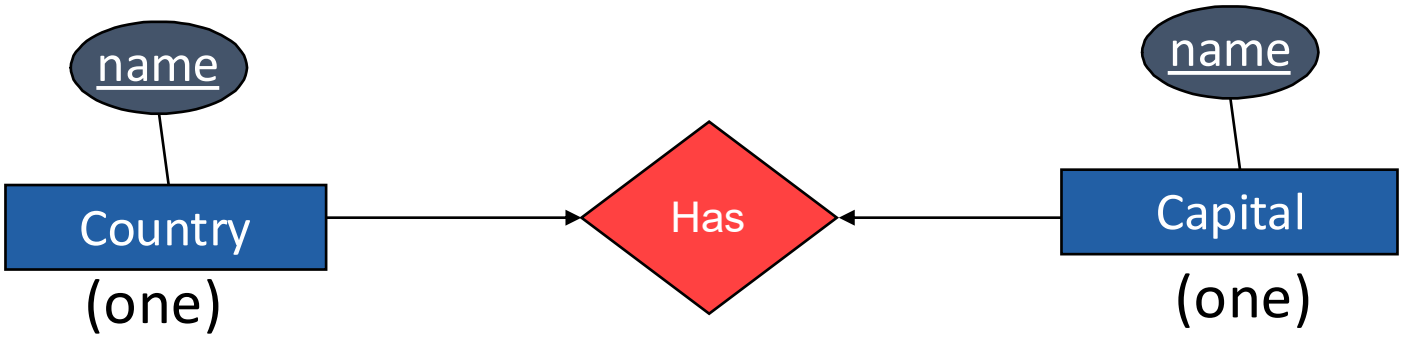
How can we identify the role of a specific MoviePerson in a specific movie

Robert Pattinson as Bruce Wayne

Clicker question: Does this allow us to represent that Robert plays both Bruce Wayne and Batman in the "The Batman"?

A. Yes. B. No.

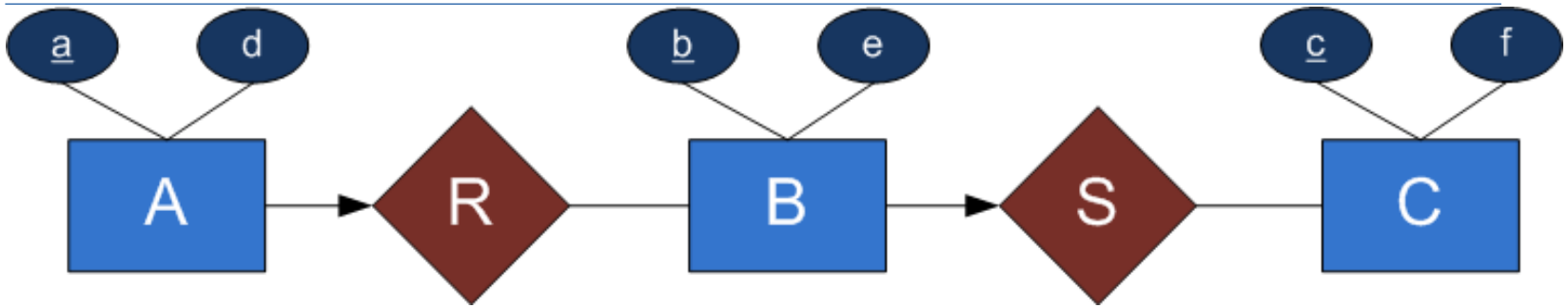
How can we uniquely identify a relationship?



Ottawa

The key of a one-to-one relationship is the key of ONE of the entities.

Clicker Question



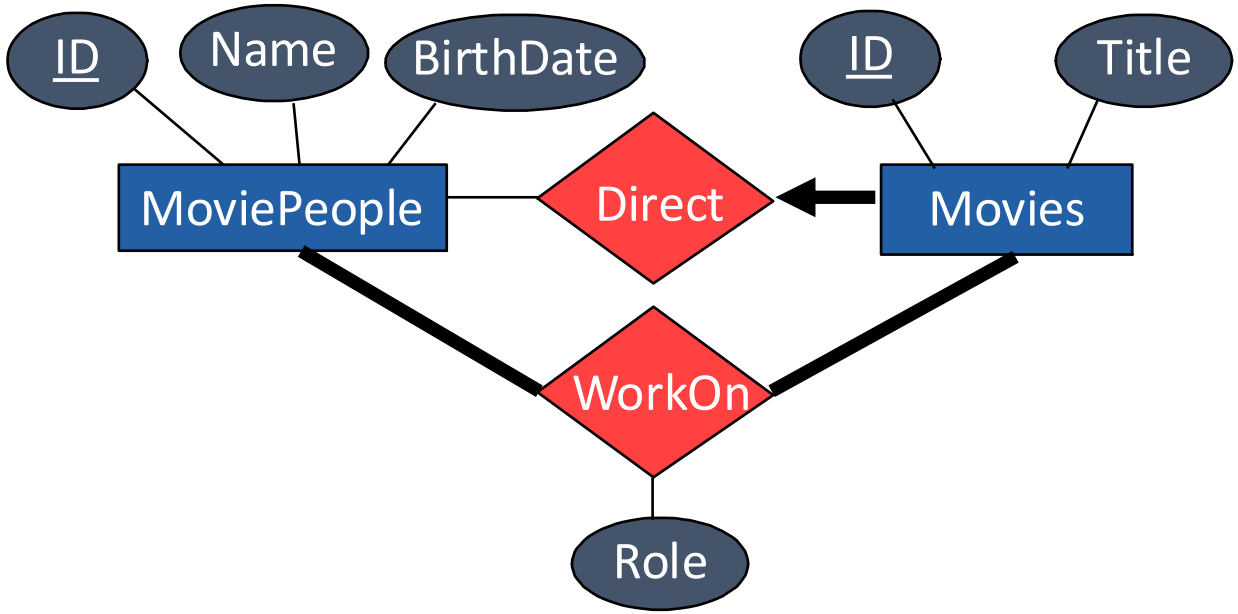
Suppose that a1 and a2 are the only entities of A, b1 and b2 are the only entities of B, and c1 and c2 are the only entities of C.

Which of the following relationship sets for R and S are possible according to the diagram, where T = {(e1,f1)} means a relationship between e1 and f1 exists in relationship set T

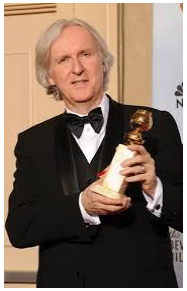
A. R = {}; S = {(b2, c1), (b2, c2)}
B. R = {}; S = {(b1, c2), (b2, c2)}
C. R = {(a2, b2)}; S = {(b2, c1), (b2, c2)}
D. R = {(a1, b2), (a2, b1), (a2, b2)}; S = {}
E. None of the above

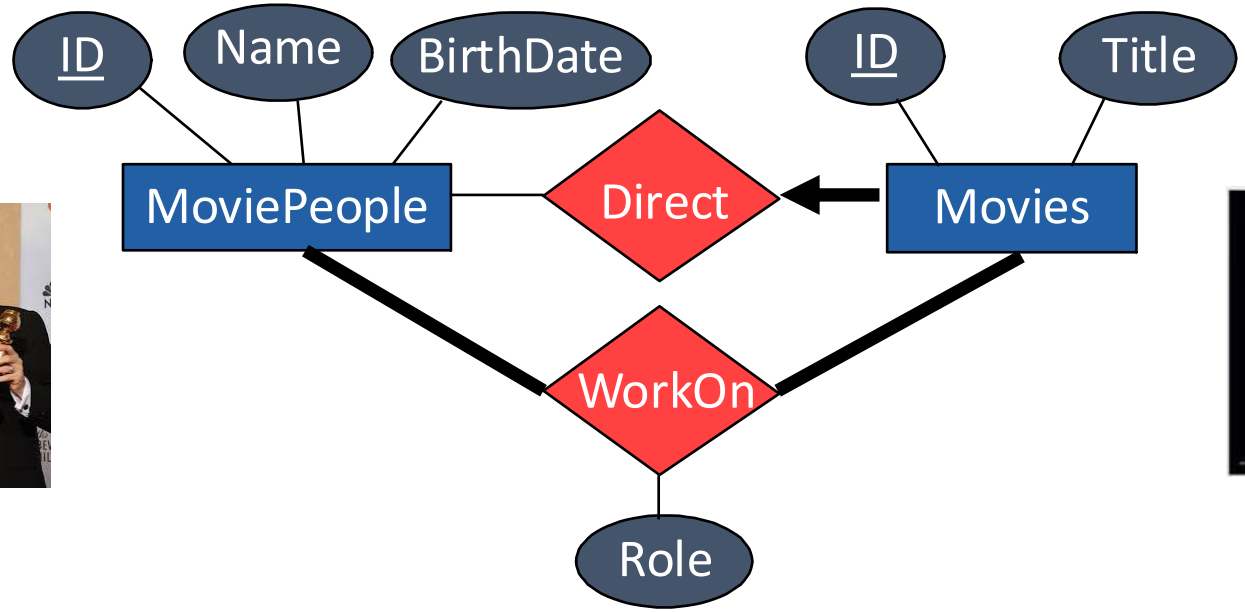
Participation Constraints

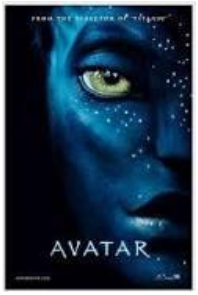
- Participation : Indicates if all entities participate in the relationship.
- An entity's participation can be ***total*** or ***partial***.
- Requiring total participation is a ***participation constraint*** and it is shown with a thick line
  - Important when considering deletions
  - i.e., participation of Movie in Directs is total (thick line)
    - Every movie must appear in some relationship in the Directs set



Why are participation constraints important

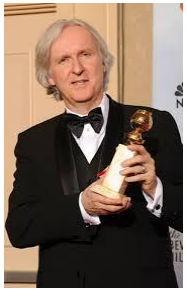


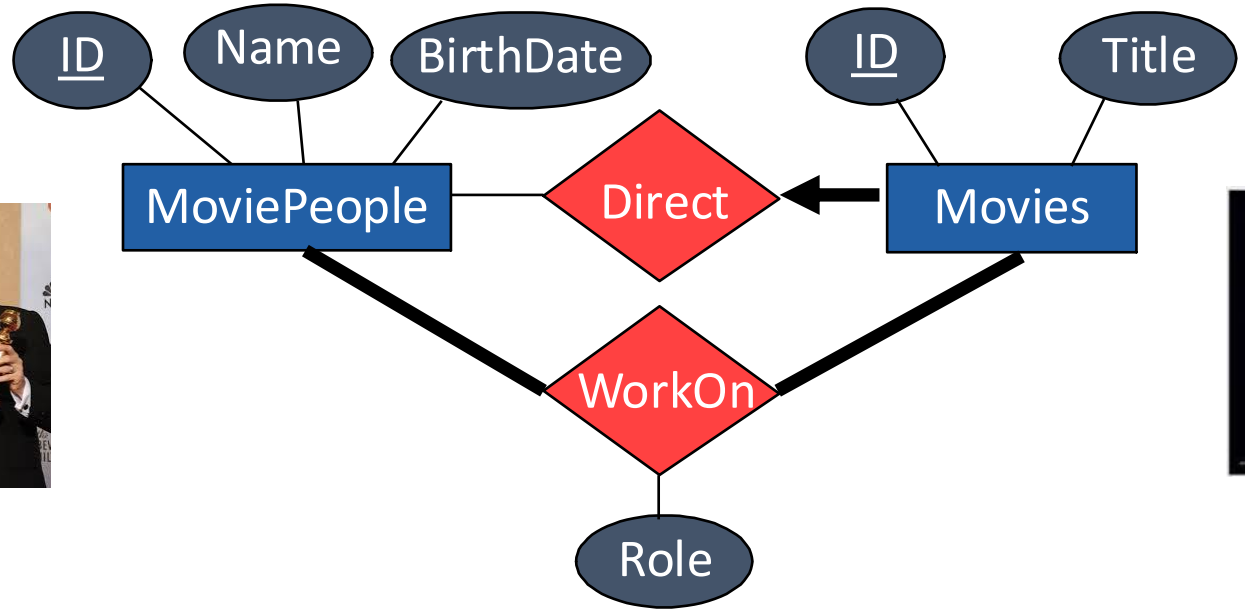


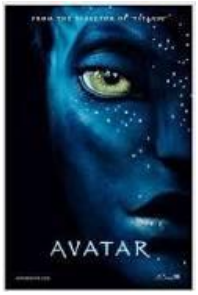


Would I be able to delete James Cameron without deleting Avatar? A. Yes B. No C. It depends

Why are participation constraints important







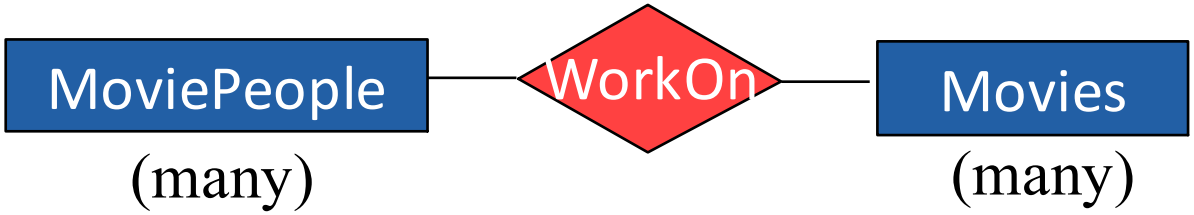
Would I be able to delete James Cameron without deleting Avatar? A. Yes B. No C. It depends

Would I be able to delete Avatar without deleting James Cameron?

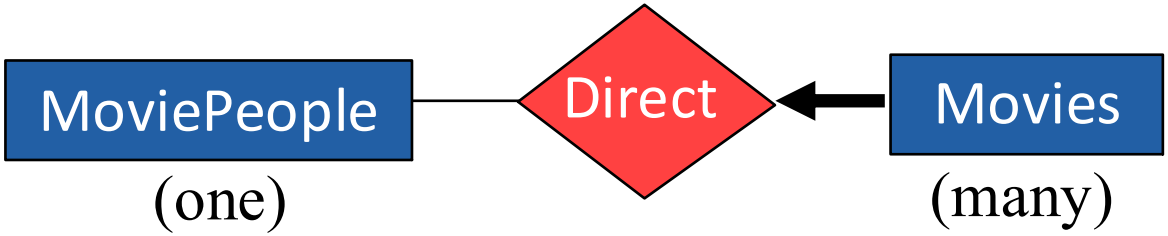
A. Yes B. No C. It depends

Line types summarized

- Plain lines mean many to many:

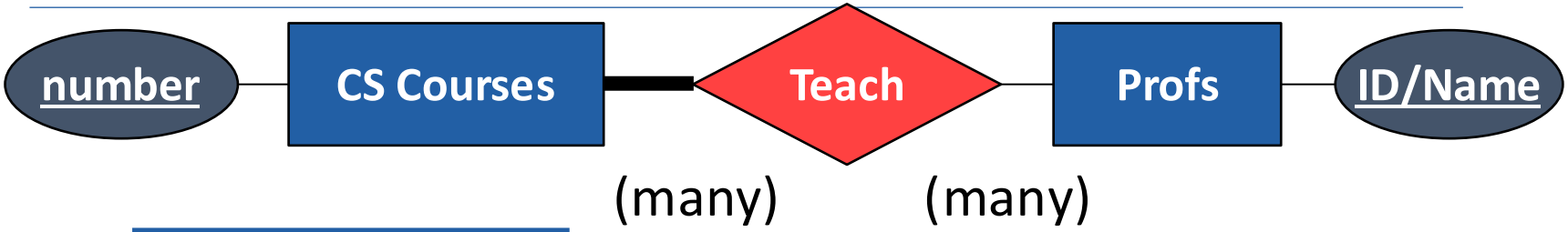


- Arrows mean the other side has a cardinality of one



- A thick line requires total participation and can be added to any line, arrow or not

Line annotation example continued
Thick lines – on one side



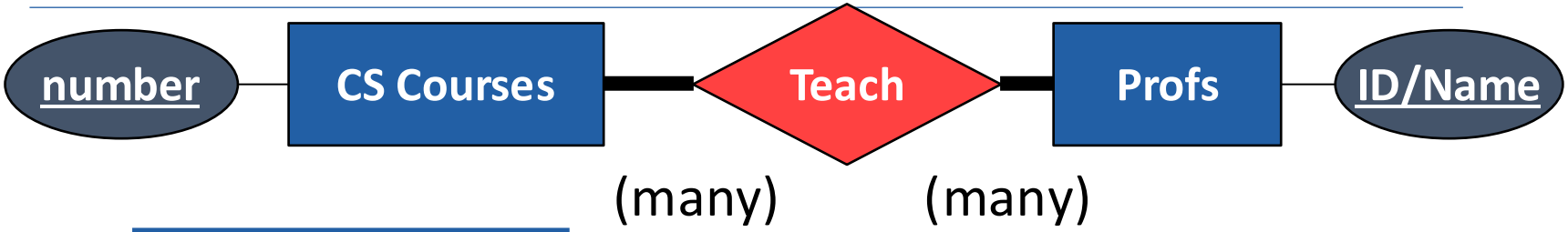
(many) (many)

CS Courses Number
100
304
312
313
417

Profs ID/Name
Rachel
Steve
Jessica
Phong

- Each course can be taught by more than one prof
- Each prof can teach more than one course
- ALL courses have to be taught by some prof
- Not all profs have to teach a course

Thick lines – on both sides



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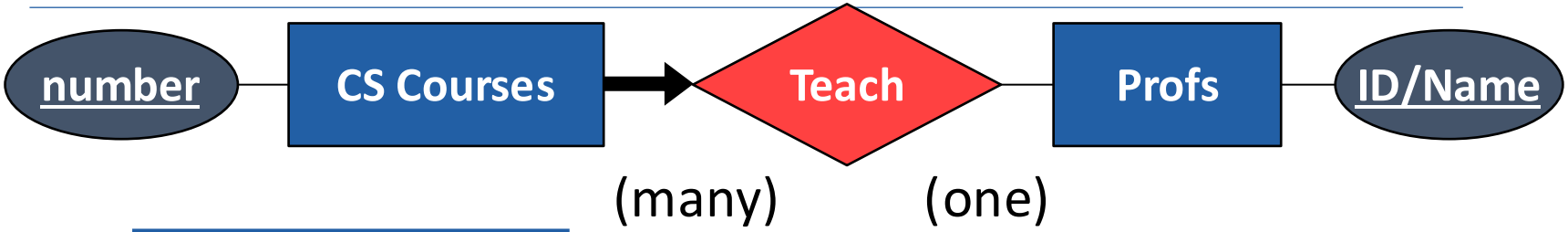
- Each course can be taught by more than one prof
- Each prof can teach more than one course
- ALL courses have to be taught by some prof
- ALL profs have to teach a course

Thick lines with arrows – on one side



(many) (one)

CS Courses Number
100
304
312
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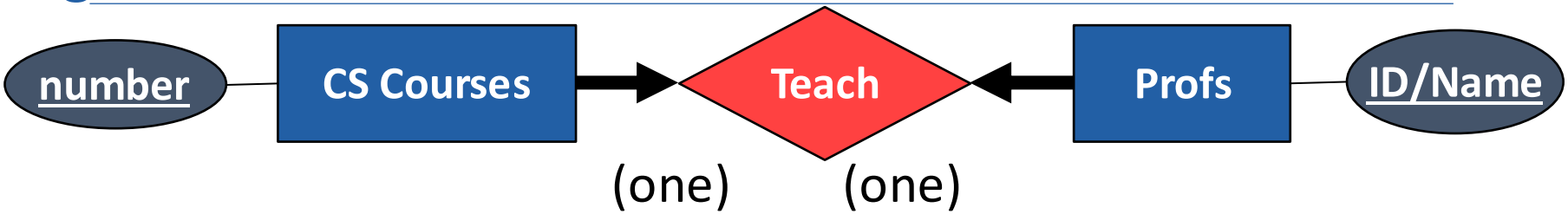
Profs ID/Name
Rachel
Steve
Jessica
Phong

- Each course can be taught ONE prof
- Each prof can teach more than one course
- ALL courses have to be taught by some prof
- Not all profs have to teach a course

Clicker question: Thick lines with arrows – on both sides: Which of the following would allow us to satisfy the given constraint?

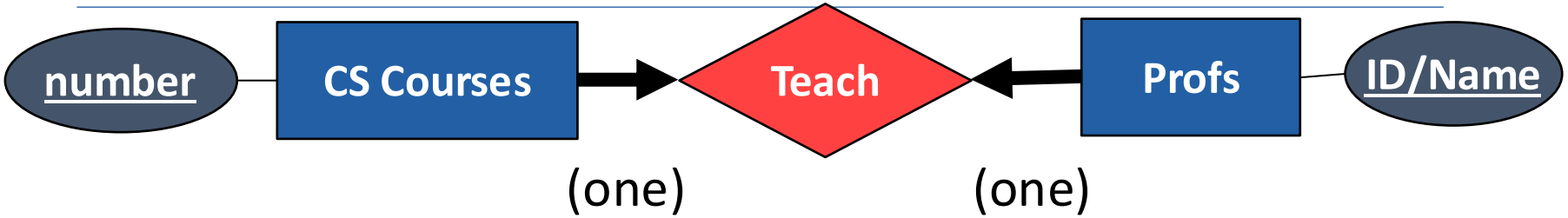


CS Courses Number
100
304
312
313
417

Profs ID/Name
Rachel
Steve
Jessica
Phong

A. Adding {(304,Jessica)} to Teach
B. Adding {(304, Jessica), (312, Steve)} to Teach
C. Both of the above
D. None of the above

Thick lines with arrows – on both sides

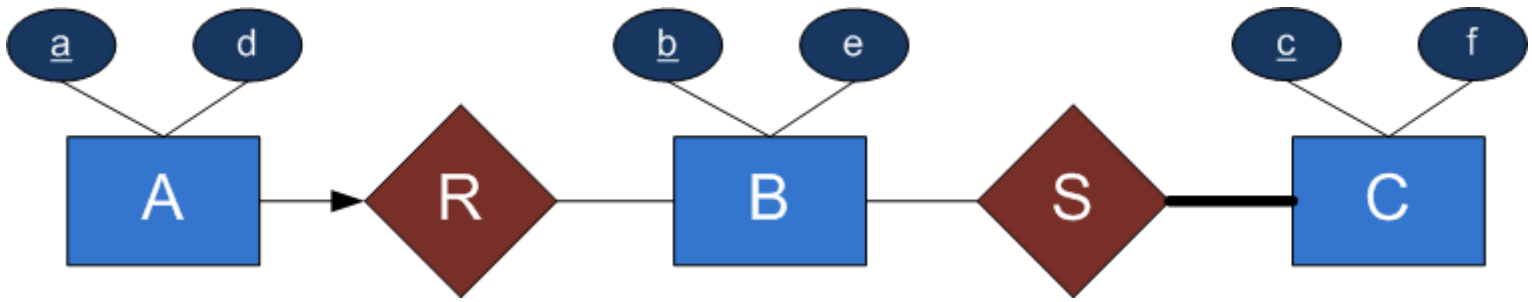


CS Courses Number
100
304
[redacted]
313
417

Profs ID/Name
Rachel
Steve
Jessica
Phong

- Each course can be taught by AT MOST one prof
- Each prof can teach AT MOST one course
- ALL courses have to be taught by some prof
- ALL profs have to teach a course
- **Some course has to go!**

Clicker Question



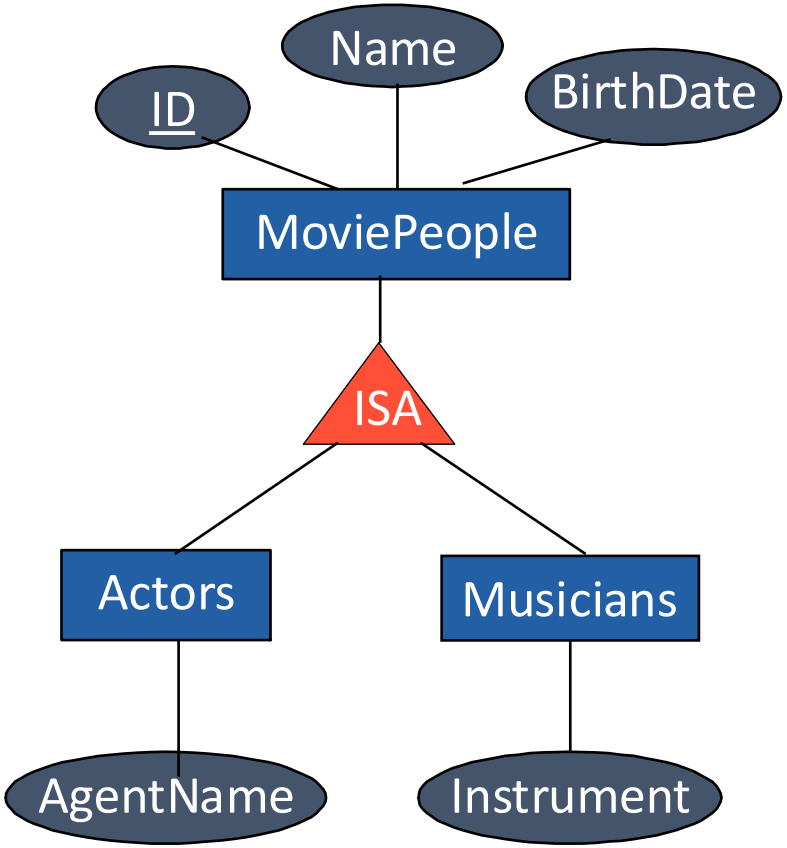
Suppose that a1 and a2 are the only entities of A, b1 and b2 are the only entities of B, and c1 and c2 are the only entities of C.

Which of the following relationship sets for R and S are possible according to the diagram, where T = {(e1,f1)} means a relationship between e1 and f1 exists in relationship set T

A. R = {}; S = {}
B. R = {(a1,b1)}, S = {(b2,c2)}
C. R = {(a1,b1), (a1,b2)}; S = {(b1,c1), (b2,c2)}
D. R = {(a1, b2)}; S = {(b1,c2), (b2, c1), (b1,c1)}
E. None of the above

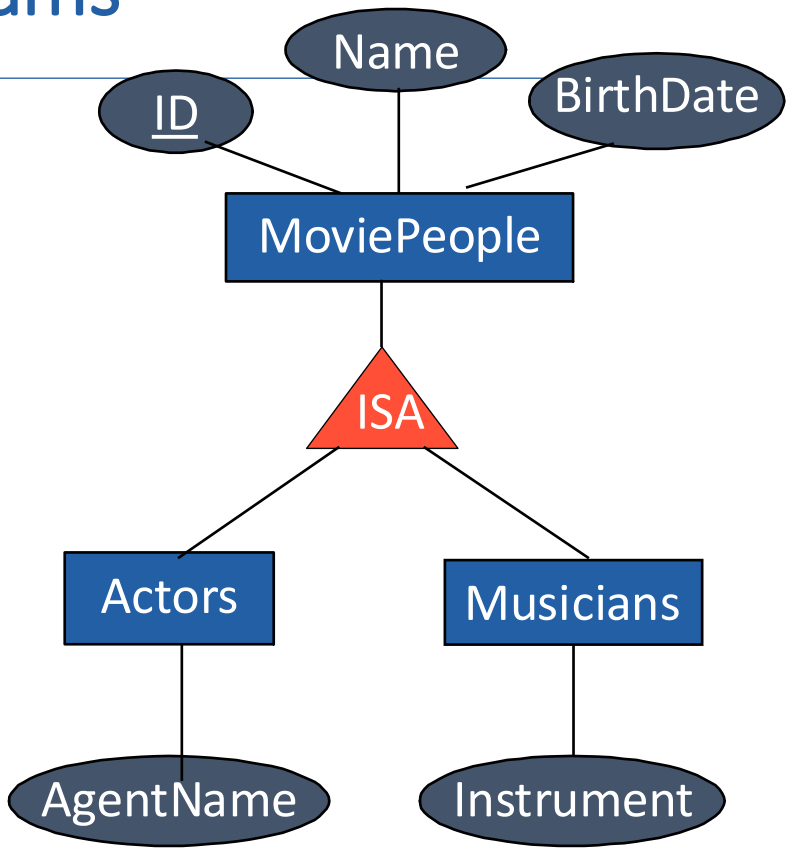
Generalization/Specialization (ISA relationships)

- As in Java, or other PLs, attributes can be inherited.
- If we declare A ***ISA*** B, every A entity is a B entity.
- Reasons for using ISA:
  - To add descriptive attributes specific to a subclass.
  - To restrict entities that participate in a relationship.



There are some ISA constraints we can't express in ER diagrams

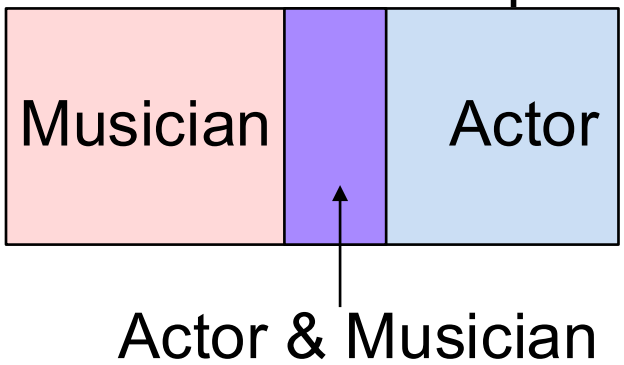
- ***Overlap constraints*** can be:
  - ***Disjoint :*** a superclass entity belongs to no more than a single subclass
  - ***Overlapping :*** subclasses may overlap
- ***Covering constraints*** can be:
  - ***Total :*** a superclass entity must belong to some subclass
  - ***Partial :*** some superclass entity may not be in any subclass



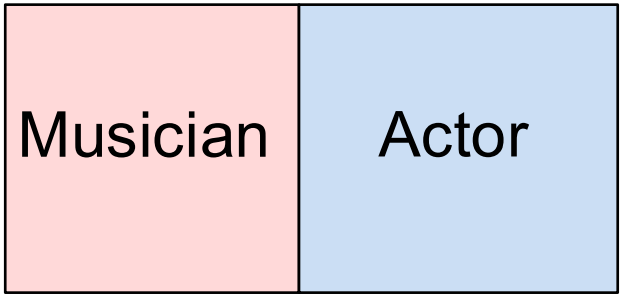
We can represent these constraints by just writing them in.

ISA constraints Illustrated

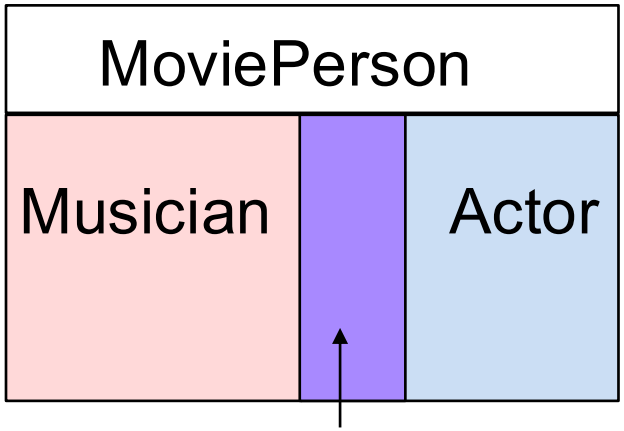
Total + Overlap



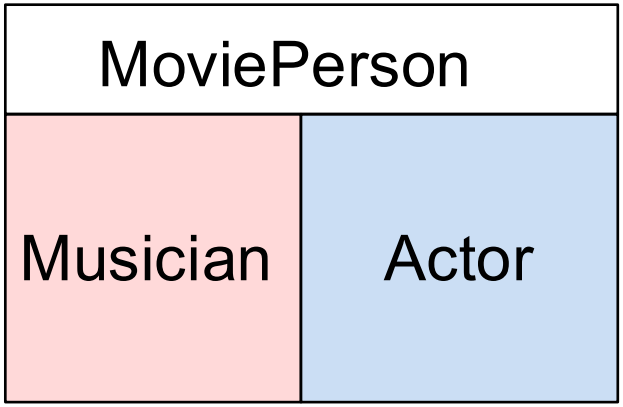
Total + Disjoint



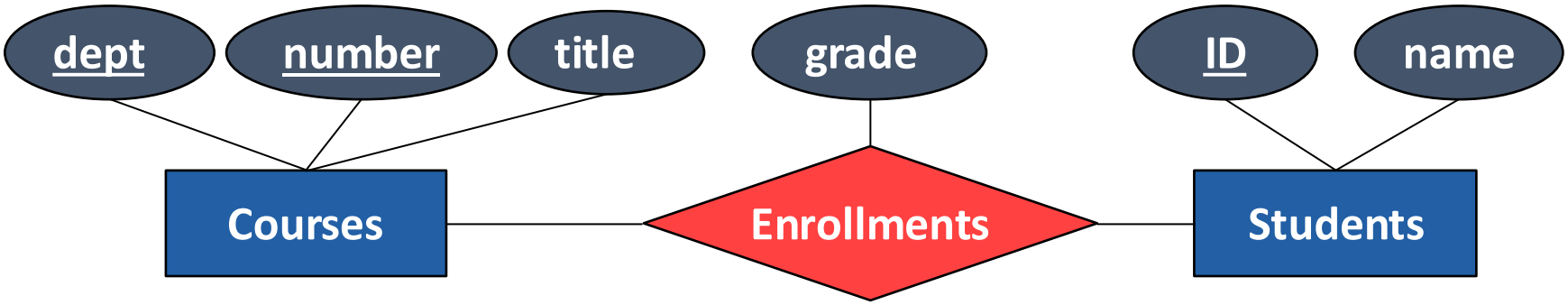
Partial + Overlap



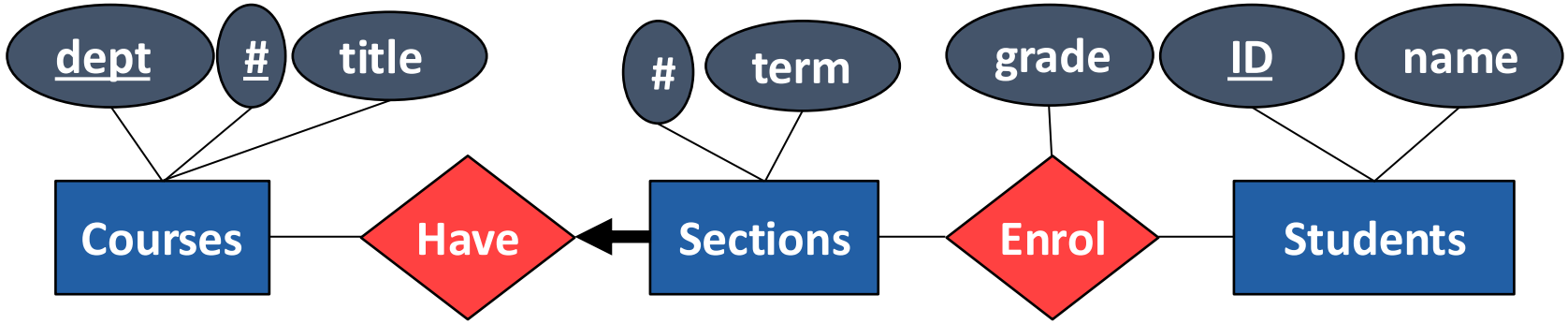
Partial + Disjoint



In a previous clicker question, we had a problem

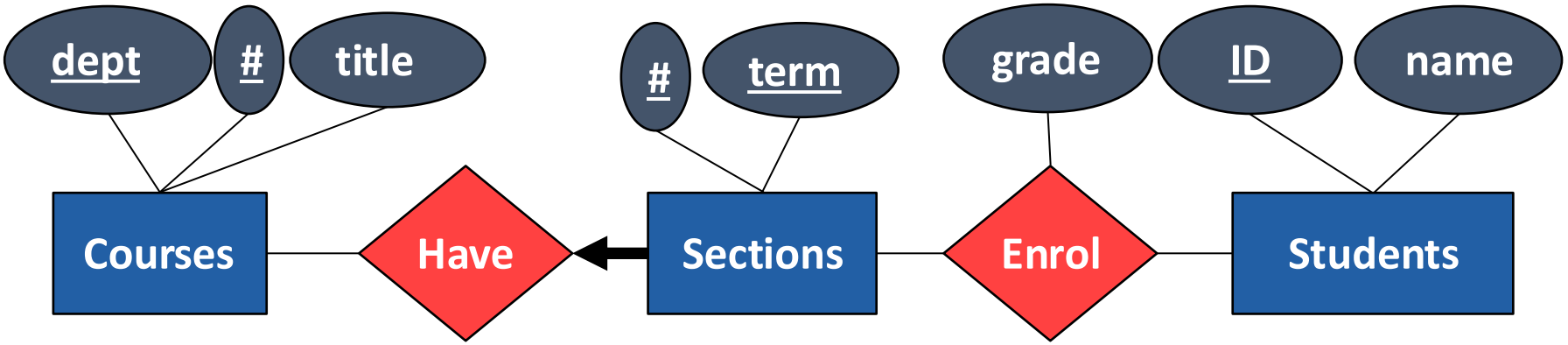


But then a student couldn't take a class more than once.
Next try:



But what should the key of Sections be?

A better solution would be if Sections could depend on Courses



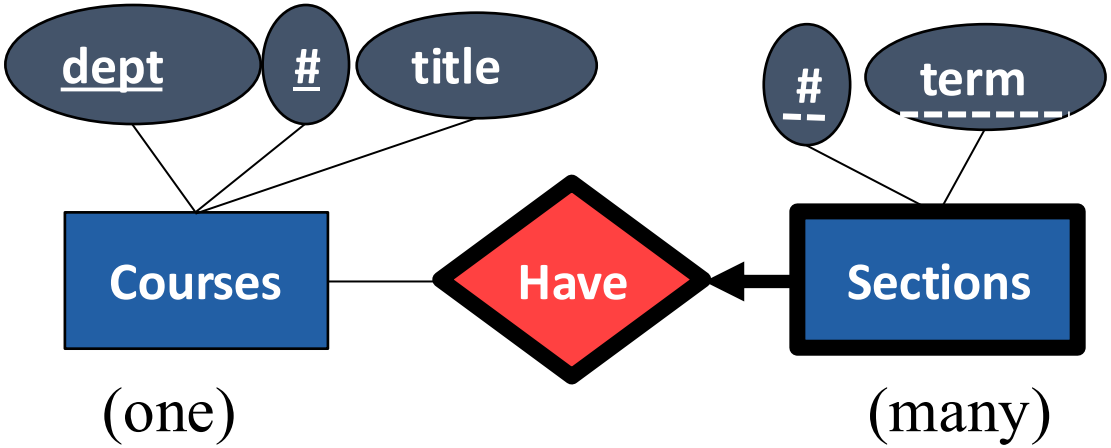
Logically, what should the key of Sections be?

- Looking only at Sections: 201 2024W2 is not unique
- Looking at Courses + Sections: CPSC 368 201 2024W2 is unique

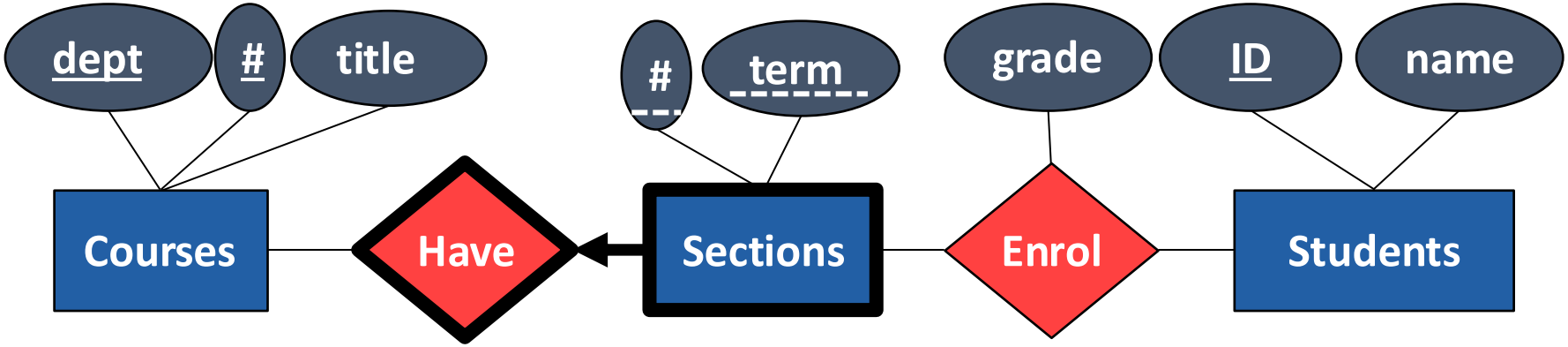
We can do this with a *weak entity*

Weak Entities

- A ***weak entity*** can be identified uniquely only by additionally considering the key of another (*owner*) entity.
  - Owner entity set and weak entity set must participate in a one-to-many relationship set (one owner, many weak entities).
  - Weak entity set must have total participation in this ***identifying*** relationship set.
  - Think of this as a "belongs to" relationship.
- Weak entity sets and their identifying relationship sets are shown with thick lines.



A better solution would be if Sections could depend on Courses



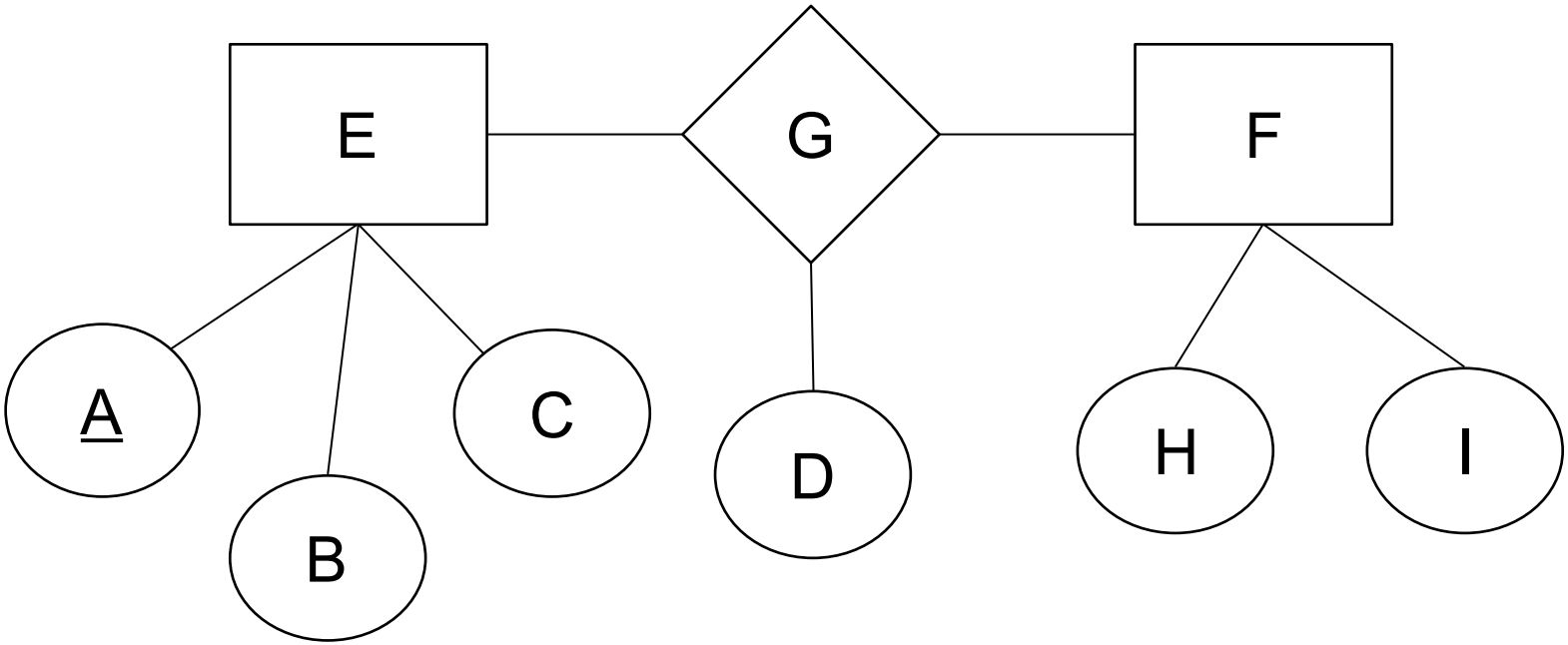
{dept, #, s#, term} is the key for Have.

{s#, term} is a partial key, underlined with a dashed line.

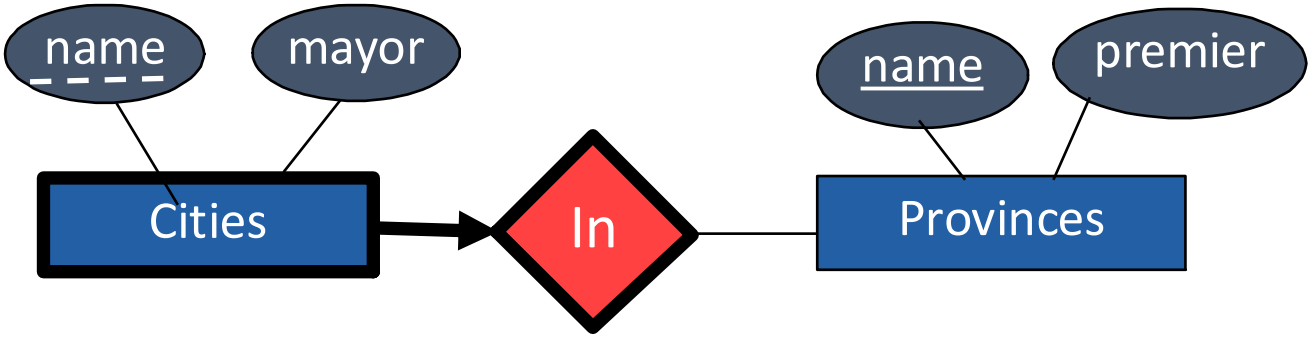
Congratulations, you can now take CPSC 368 twice (yay?)

Exercise (don't need to hand this in)

Fix the diagram below to say that F is a weak entity set with a partial key, H, and its owner entity set is E.



Clicker Question



Based on the diagram above, which of the following **must** to be true:

A. No two provinces can have premiers with the same name.

B. No two cities can have mayors with the same name.

C. No two cities can have the same name.

D. No person can be the mayor of Cities In two different provinces.

E. None of the above

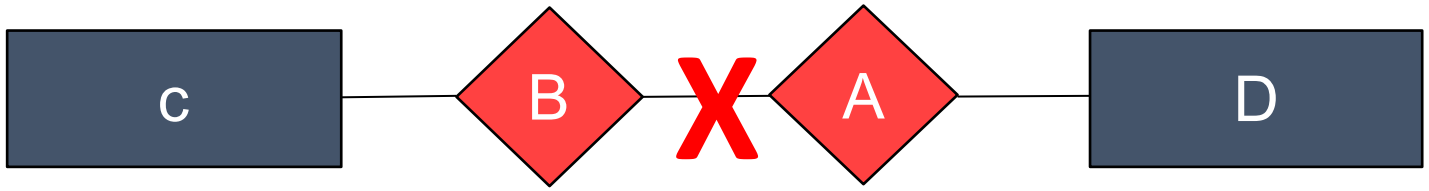
Exercise: Hockey Teams (don't need to hand this in)

- A hockey team has multiple players
- Each team has a name and a home city.
- No two teams can have the same name and home city combination.
- We do not need to retain past home city or name information.
- A player can only play for one team (we do not retain past team information), and every player must be on a team
- Each player has a unique jersey number; the combination of the jersey number and the team information is unique
- Every team must have players
- We retain the date when the player last started playing for the team.

Exercise: Hockey Teams (don’t need to hand this in)

Aggregation

- Having a relationship between relationships is forbidden.



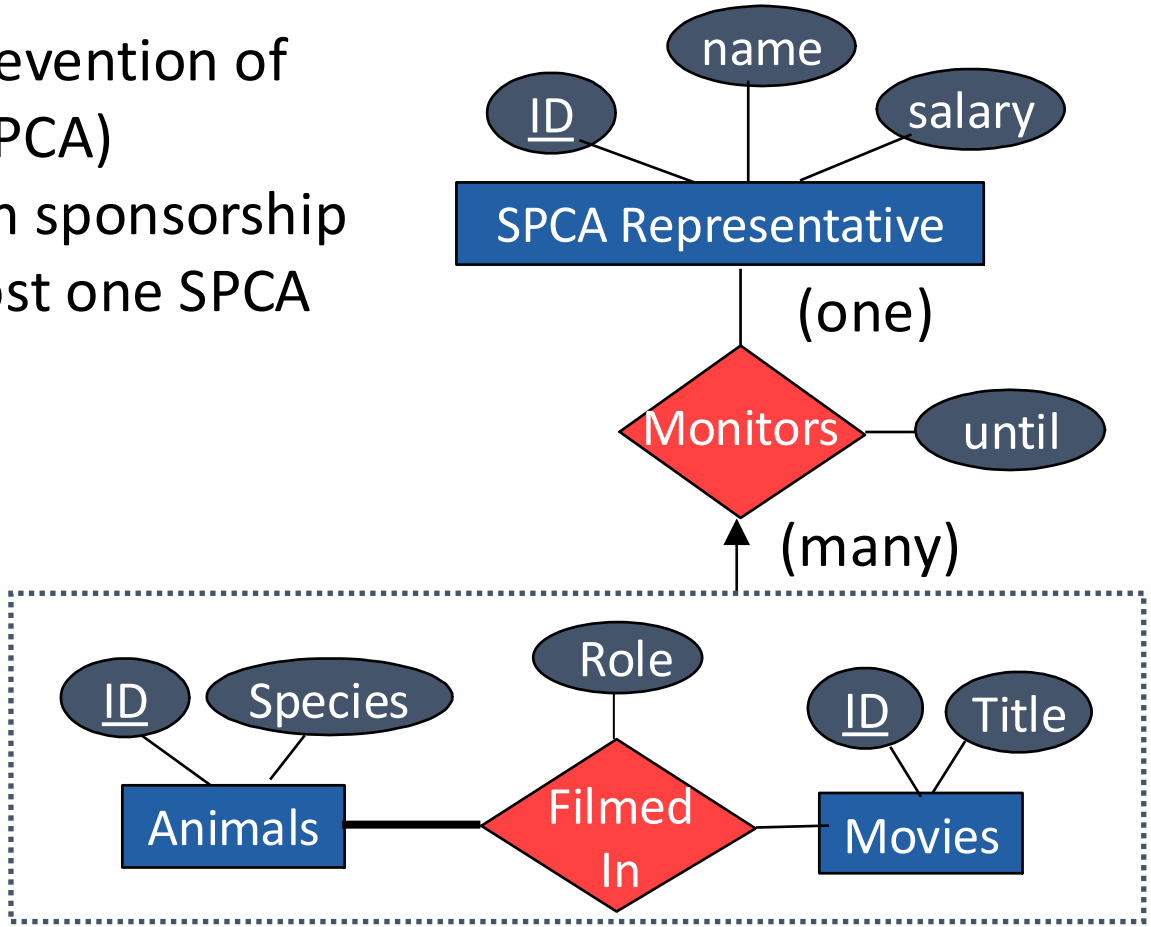
- *Aggregation* allows us to treat a relationship set as an entity set for purposes of participation in (other) relationships

Aggregation: getting around relationships between relationships

- The Society for the Prevention of Cruelty to Animals (SPCA) monitors movies. Each sponsorship is monitored by at most one SPCA representative
- What is the key for FilmedIn?

  Animal ID, movie ID
- What is the key for Monitors?

  Animal ID, movie ID



- This differs from a ternary relationship because monitors is its own relationship with a descriptive attribute

Clicker Question

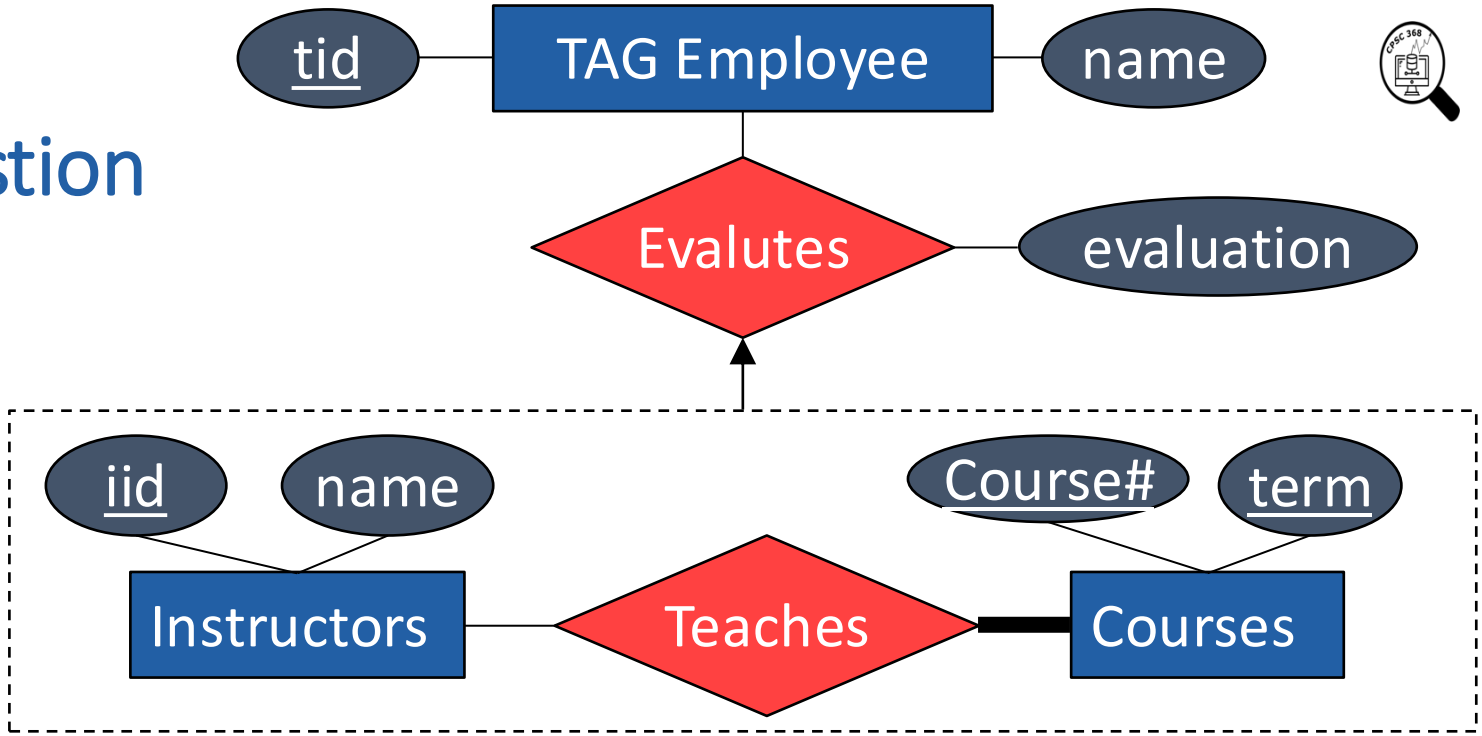


Figure out the (minimal) keys for each entity set and each relationship set in the above diagram.

Clicker Question

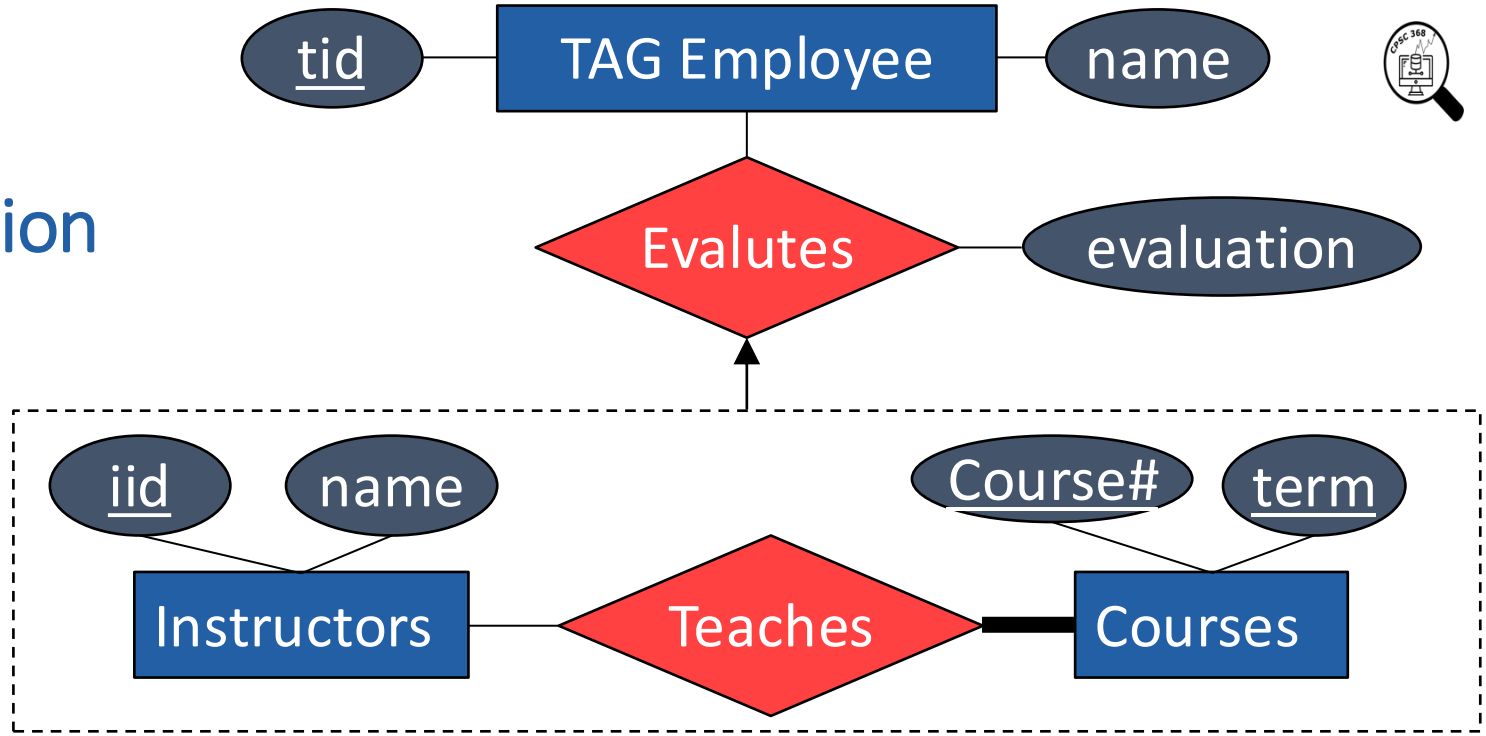


Figure out the (minimal) keys for each entity set and each relationship set in the above diagram.

Choose the correct choice of (minimal) key from the options below:

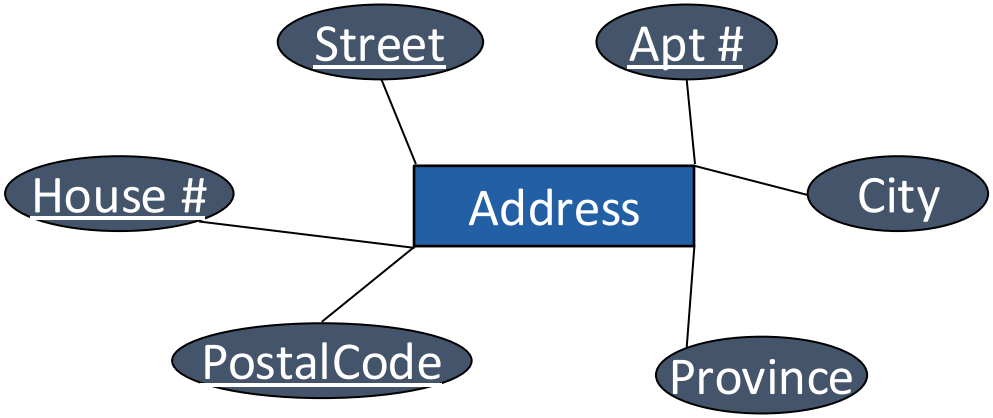
A. The (minimal) key of Evaluates is tid
B. The (minimal) key of Evaluates is iid + course# + term.
C. The (minimal) key of Evaluates is iid + course# + term + tid
D. The (minimal) key of Evaluates is iid+ course# + term + evaluation
E. None of the above

Summary

Name	Symbol
Entity	Entity
Attribute	Attribute
Relationship	Relationship
Generalization/Specialization	IsA
Weak Entity	Weak Entity
Aggregation	
Participation Constraints	
key constraints	

That's all there is to it

- Some ER models differ in expressiveness
- They model *most* concepts people want
- They don't model all of them, e.g.,
  - Functional dependencies – some attributes determine some other attributes, e.g., postal code determines (only) city and province



Conceptual Design Using the ER Model

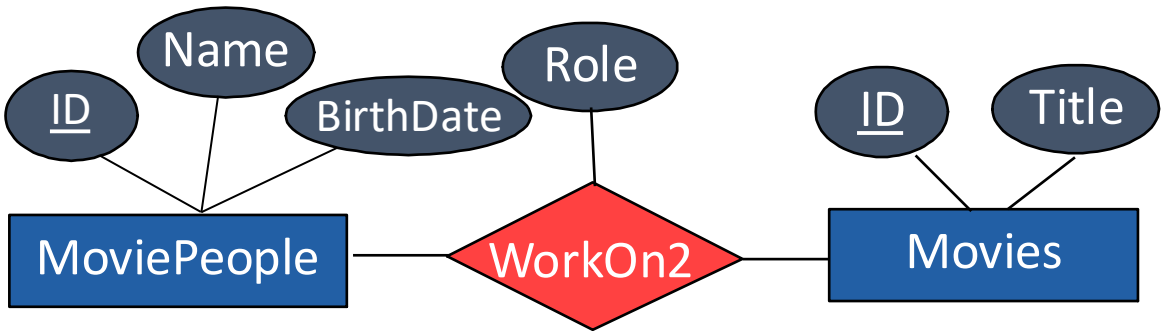
- Design choices:
  - Should a concept be modeled as an entity or an attribute?
  - Should a concept be modeled as an entity or a relationship?
  - Identifying relationships: Binary or ternary? Aggregation?
- Constraints in the ER Model:
  - A lot of data semantics can (and should) be captured.
  - But some constraints cannot be captured in ER diagrams.
    - i.e. domain constraints
    - dependencies

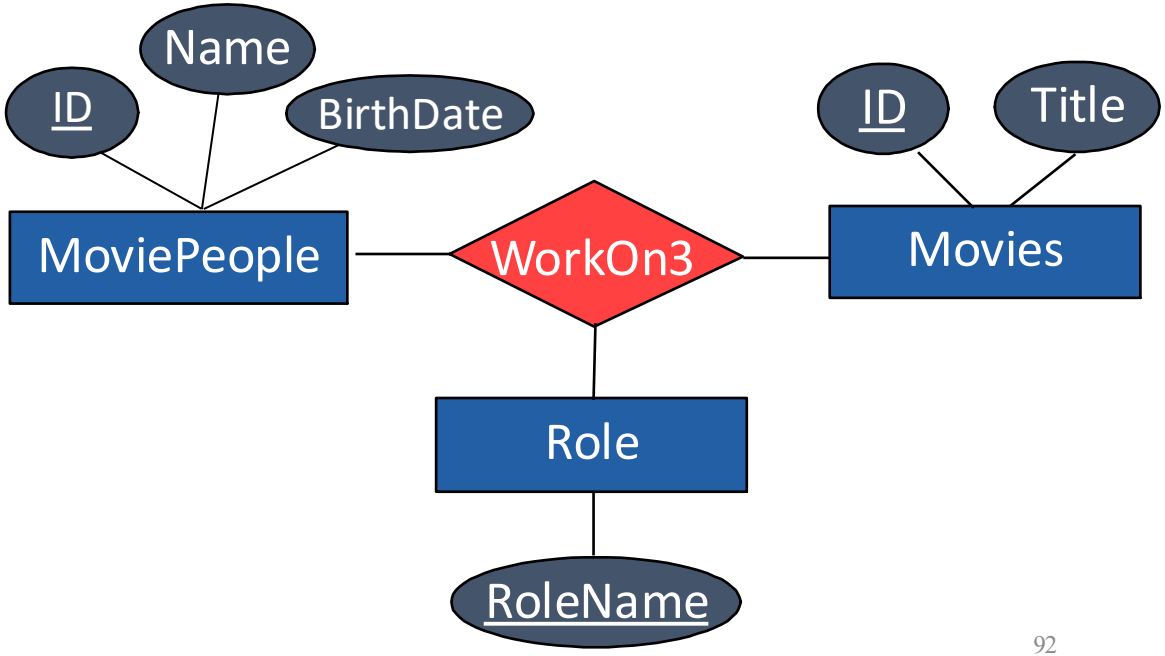
Entity vs. Attribute

- Should an *address* be an attribute of MoviePeople or an entity (connected to MoviePeople by a relationship)?
- Depends upon
  - the use we want to make of address information
  - the semantics of the data:
    - If we have several addresses per person, *address* must be an entity (since attributes cannot be set-valued).
    - If a person has only one street address, one city, one province, one postal code, etc. then these should simply be attributes.

Entity vs. Attribute (Cont.)

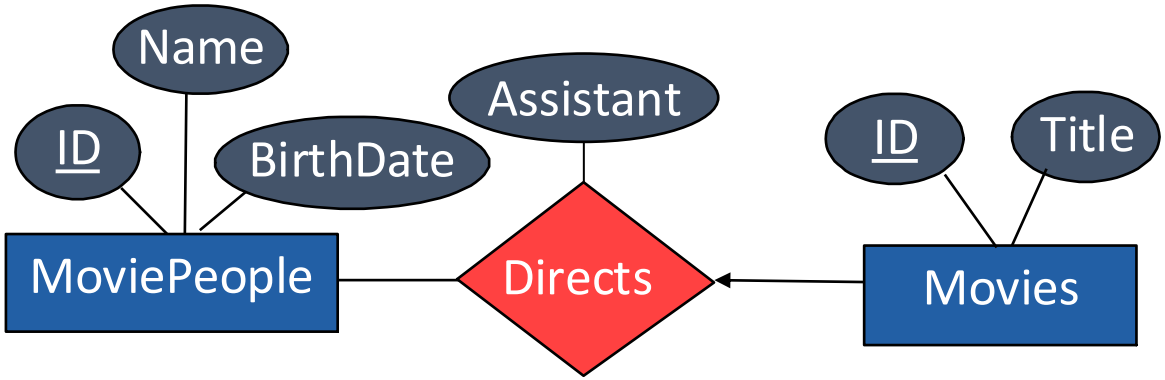
- WorkOn2 does not allow a person to have more than one role in the same movie.
- We want to associate the same pair (MoviePerson, Movie) with more than one set of values for the descriptive attributes?
- Solution: change descriptive attributes into entities.

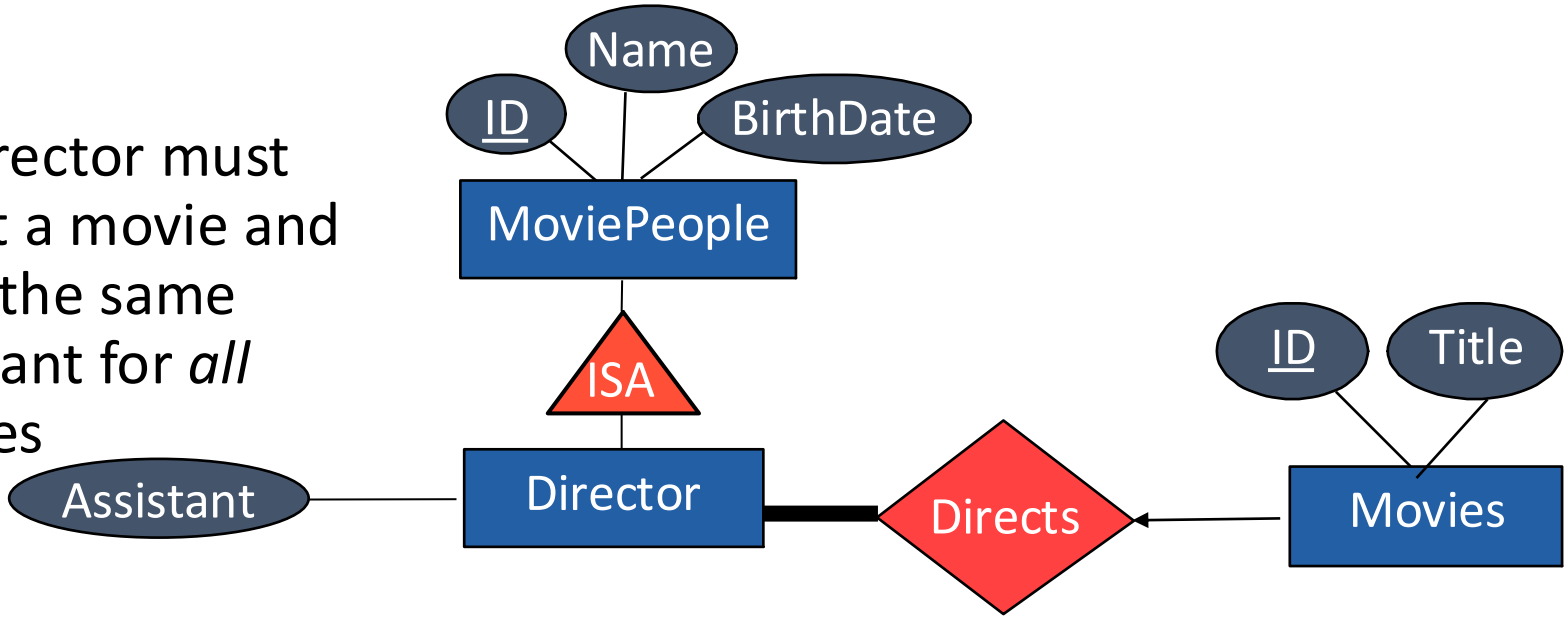




Entity vs. Relationship

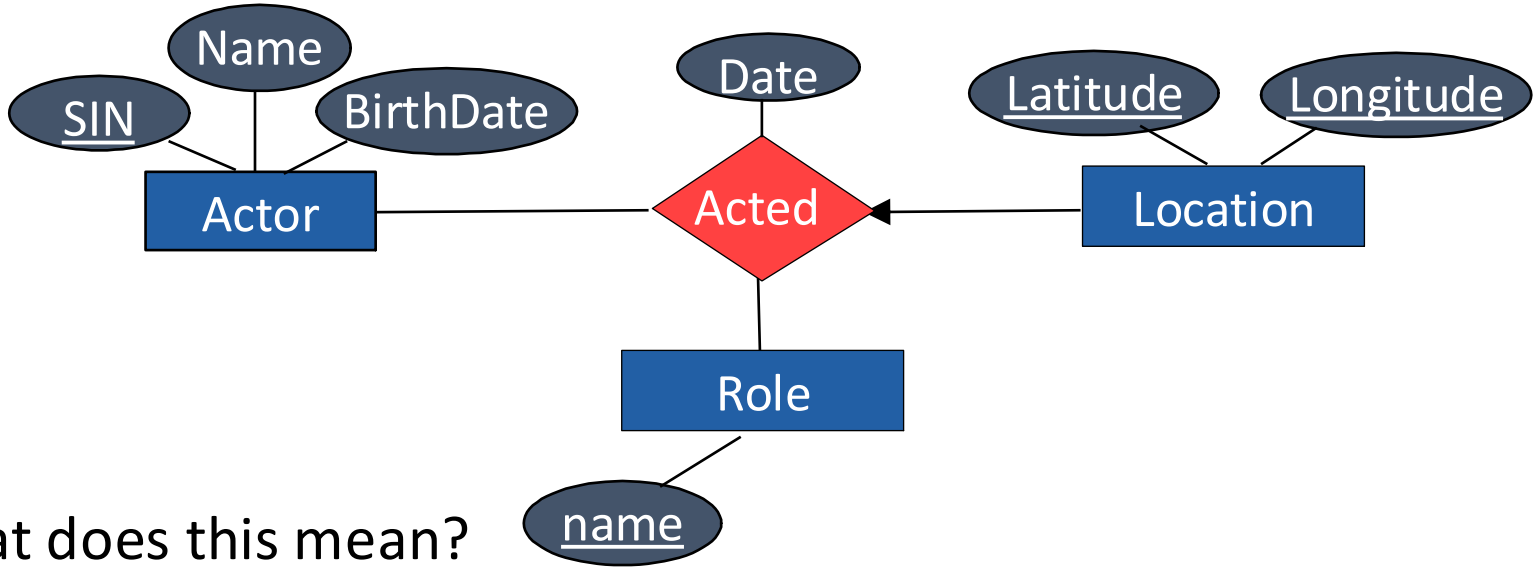
- How are the two ER models different?
- Director can get a separate assistant for each movie.
- All director must direct a movie and have the same assistant for *all* movies





Key constraints on non-binary relationships

Ternary relation Acted relates entity sets Location, Actor and Role, and has descriptive attribute *date*.



What does this mean?

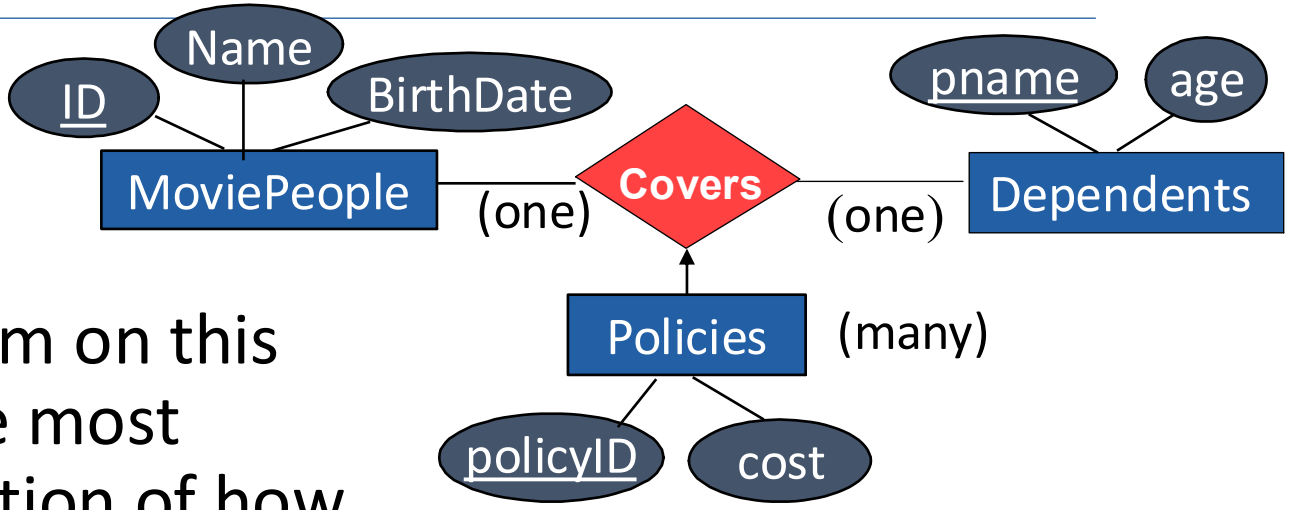
A. If I know the location, then I know both the actor and the role

B. If I know Location, I only know the role

C. If I know Location, I only know the actor

D. Other. I don't know. My head hurts. Make it stop.

Ternary relationship constraint understanding: clicker question



Given the diagram on this slide, what is the most accurate description of how many dependents can one policy cover

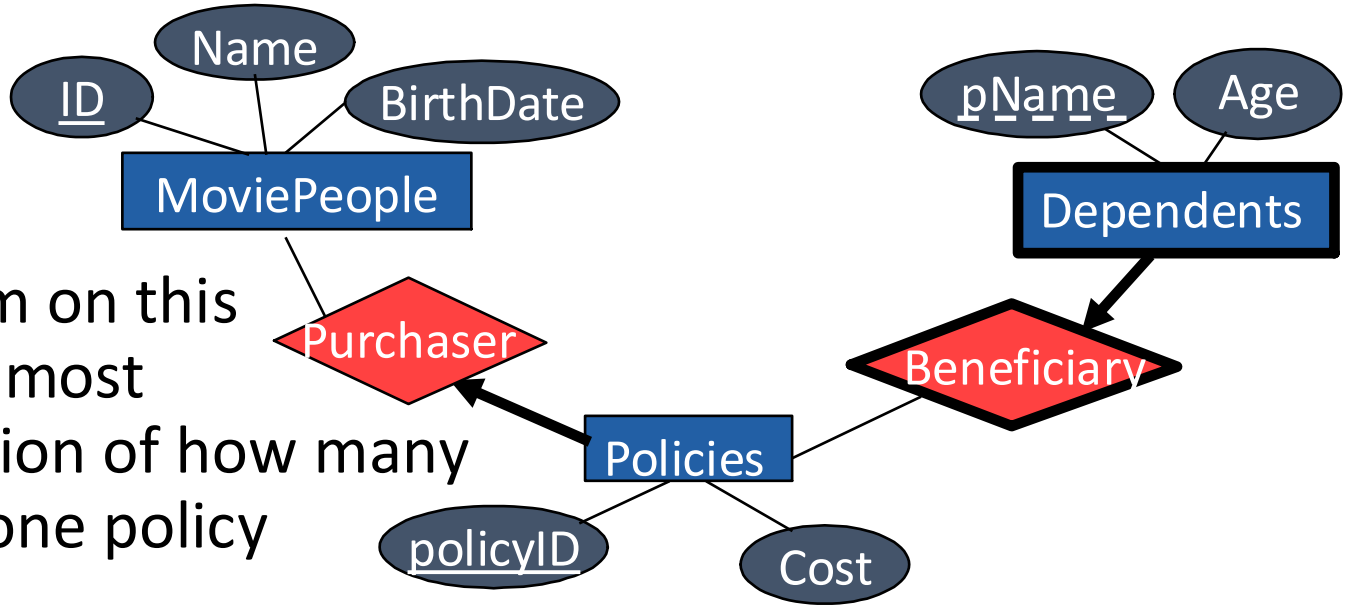
A. At least one

B. At most one

C. Exactly one

D. There are no restrictions

Comparing Binary and Ternary Relationships



Given the diagram on this slide, what is the most accurate description of how many dependents can one policy cover?

A. At least one

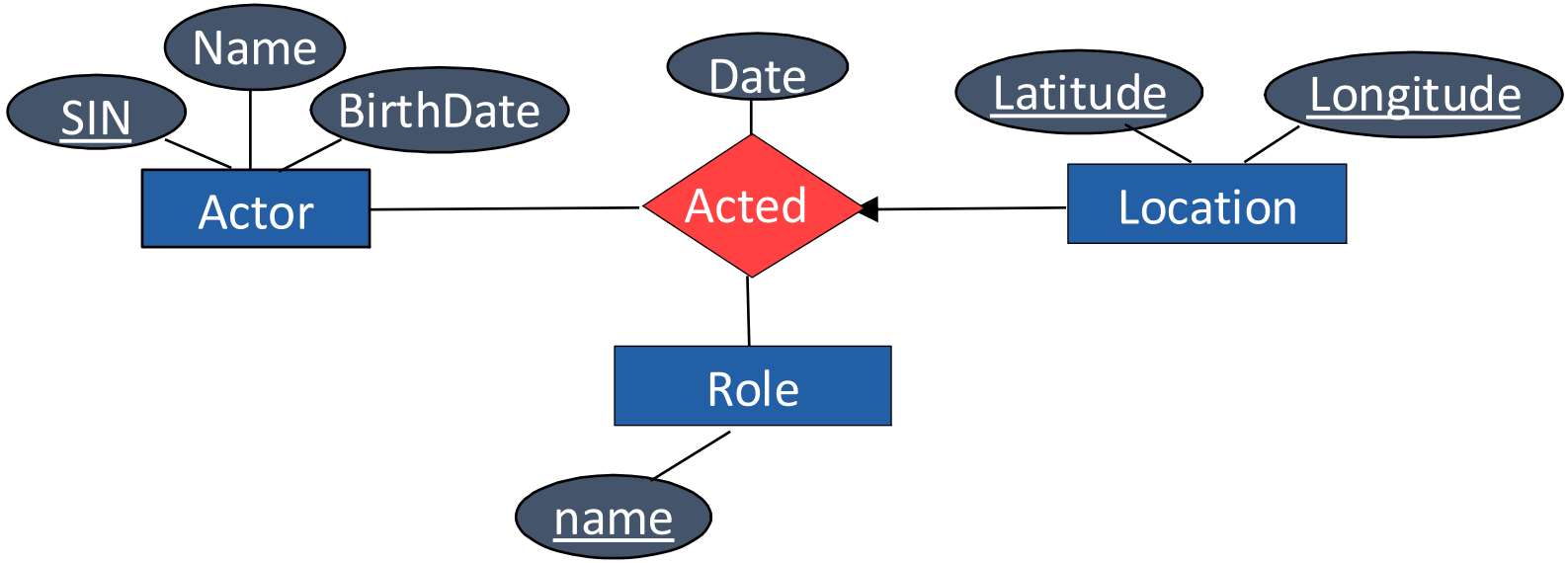
B. At most one

C. Exactly one

D. There are no restrictions

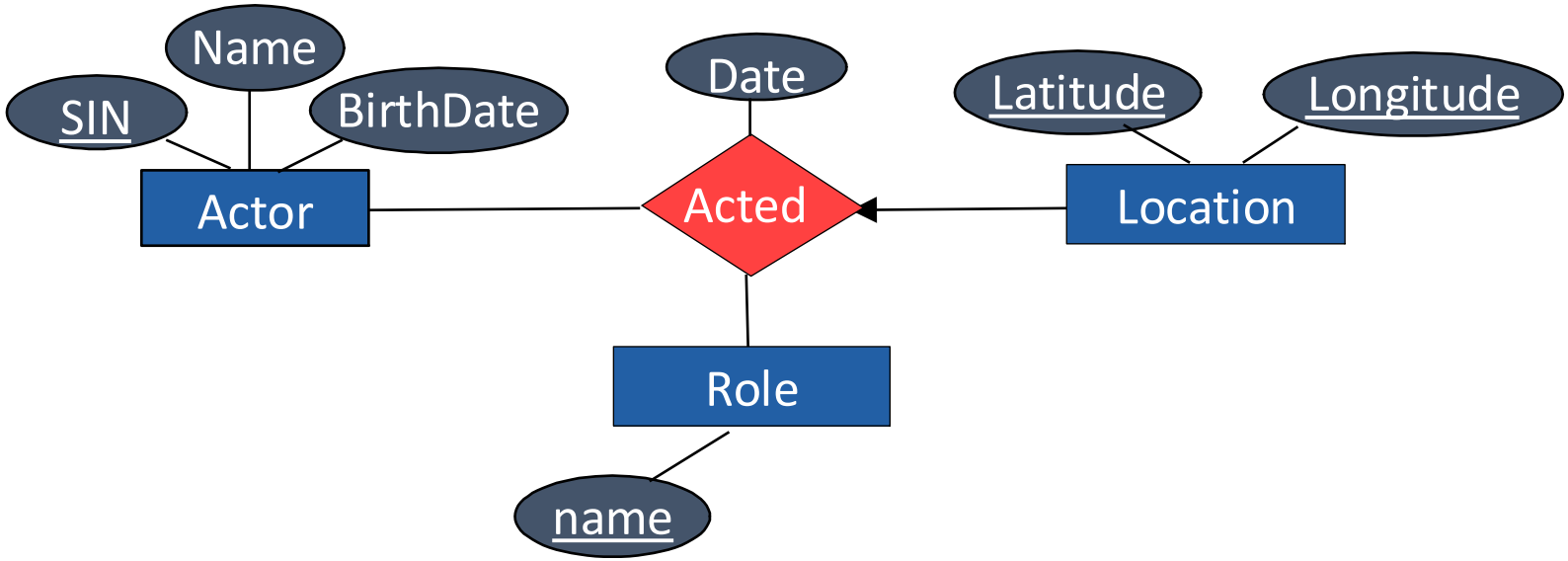
Binary vs. Ternary Relationships

- An example in the other direction: a ternary relation Acted relates entity sets Actor, Location, and Role, and has descriptive attribute *date*.



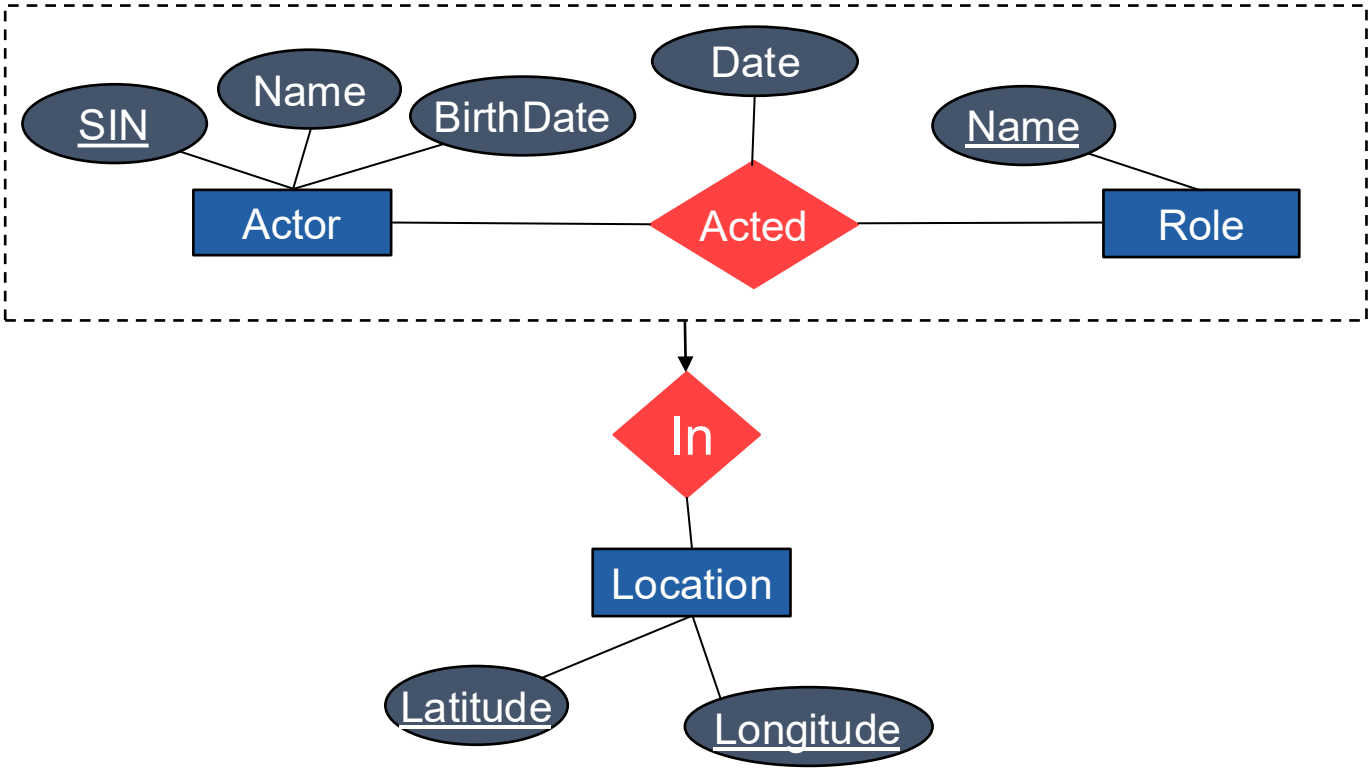
- Can we use two of binary relationships instead?

Binary vs. Ternary Relationships vs. Aggregation



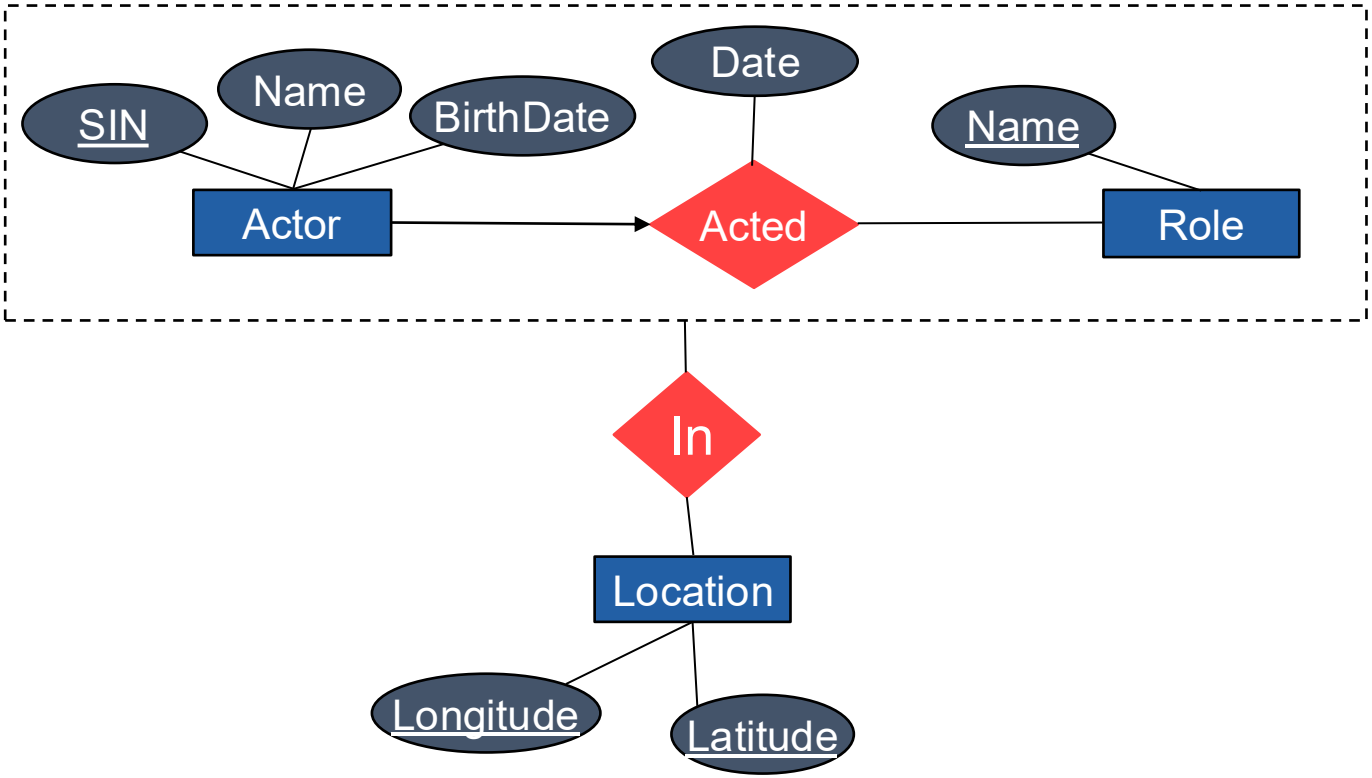
- No combination of binary relationships is an adequate substitute:
  - A "acted as" R, R "visited" L, and A "Acted in" L : does not imply that A acted as R in L.
  - Also, how would we record *date*?

Binary vs. Ternary Relationships vs. Aggregation



- Aggregation can be used instead of a ternary relation if need to impose additional constraints:
  - I.e., If you know the person and the role, you know where they acted.

Binary vs. Ternary Relationships vs. Aggregation



- Aggregation can be used instead of a ternary relation if need to impose additional constraints:
  - I.e. A person cannot act in more than one role but they can film in many different locations.

Summary of Conceptual Design

- *Conceptual design* follows *requirements analysis*,
  - Yields a high-level description of data to be stored
- ER model popular for conceptual design
  - Constructs are expressive, close to the way people think about their applications.
- Basic constructs: *entities*, *relationships*, and *attributes* (of entities and relationships).
- Some additional constructs: *weak entities*, *ISA relationships*, and *aggregation*.
- Note: There are many variations on ER model.

Summary of ER (Cont.)

- Several kinds of integrity constraints can be expressed in the ER model: *key constraints*, *participation constraints*, and *overlap/covering constraints* for ISA relationships. Some *foreign key constraints* are also implicit in the definition of a relationship set.
  - Some constraints (notably, *functional dependencies*) cannot be expressed in the ER model.
  - Constraints play an important role in determining the best database design for an enterprise.

Summary of ER (Cont.)

- ER design is *subjective*. There are often many ways to model a given scenario! Analyzing alternatives can be tricky, especially for a large enterprise. Common choices include:
  - entity vs. attribute
  - entity vs. relationship
  - binary or n-ary relationship
  - whether or not to use ISA hierarchies
  - whether or not to use aggregation
- Ensuring good database design: resulting relational schema should be analyzed and refined further.

Learning Goals revisited

- Explain the purpose of an ER diagram, and list the major components.
- Given a problem description, create an ER diagram given a specification. Justify the decisions you make for entities, relationships, keys, key constraints, participation constraints, weak entities, is-a relationships, and aggregations.
- given a problem description, identify alternative representations of the problem concepts and evaluate the choices
- compare alternative ER models for the same domain and identify their strengths and weaknesses

Forming Groups

- Everyone has a particular working style and preference
- Make sure that this works with your group before you commit to being a group together
- Be honest about the amount of work you want to do
  - It's not fair to anyone if things start off with different expectations (it's also super stressful and makes everyone very unhappy)